

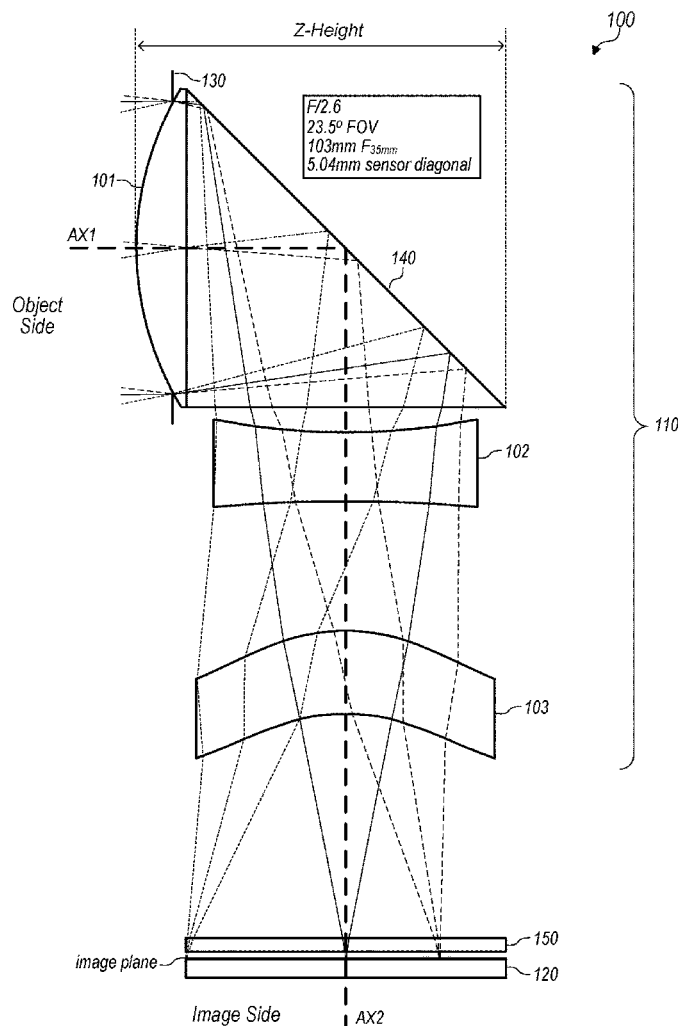


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(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2022/0050277 A1**
Yao et al. (43) **Pub. Date: Feb. 17, 2022**(54) **FOLDED LENS SYSTEM WITH THREE REFRACTIVE LENSES****Publication Classification**(51) **Int. Cl.**
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(52) **U.S. Cl.**
CPC G02B 13/0065 (2013.01); **G02B 13/0035** (2013.01)(71) Applicant: **Apple Inc.**, Cupertino, CA (US)
(72) Inventors: **Yuhong Yao**, San Jose, CA (US);
Yoshikazu Shinohara, Cupertino, CA (US)(73) Assignee: **Apple Inc.**, Cupertino, CA (US)(57) **ABSTRACT**(21) Appl. No.: **17/515,072**(22) Filed: **Oct. 29, 2021****Related U.S. Application Data**

- (63) Continuation of application No. 16/593,888, filed on Oct. 4, 2019, now Pat. No. 11,163,141, which is a continuation of application No. 15/472,138, filed on Mar. 28, 2017, now Pat. No. 10,437,023.
- (60) Provisional application No. 62/314,350, filed on Mar. 28, 2016, provisional application No. 62/334,403, filed on May 10, 2016.

Compact folded lens systems are described that may be used in small form factor cameras. Lens systems are described that may include three lens elements with refractive power, with a light folding element such as a prism, located between a first lens element on the object side of the lens system and a second lens element, that redirects the light refracted from the first lens element from a first axis onto a second axis on which the other lens elements and a photosensor are arranged. The lens systems may include an aperture stop located behind the front vertex of the lens system, for example at the first lens element, and an optional infrared filter, for example located between the last lens element and a photosensor.



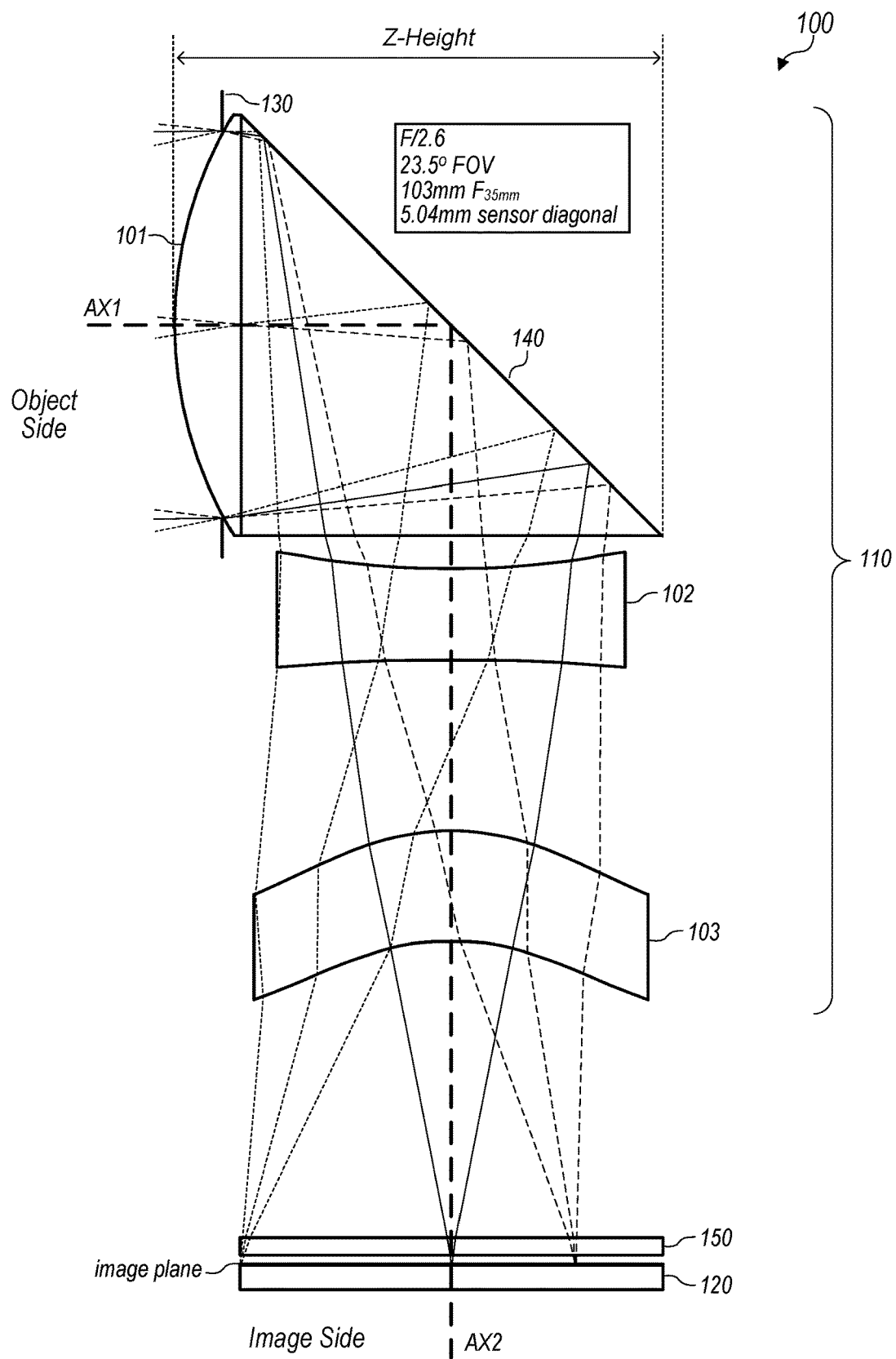


FIG. 1A

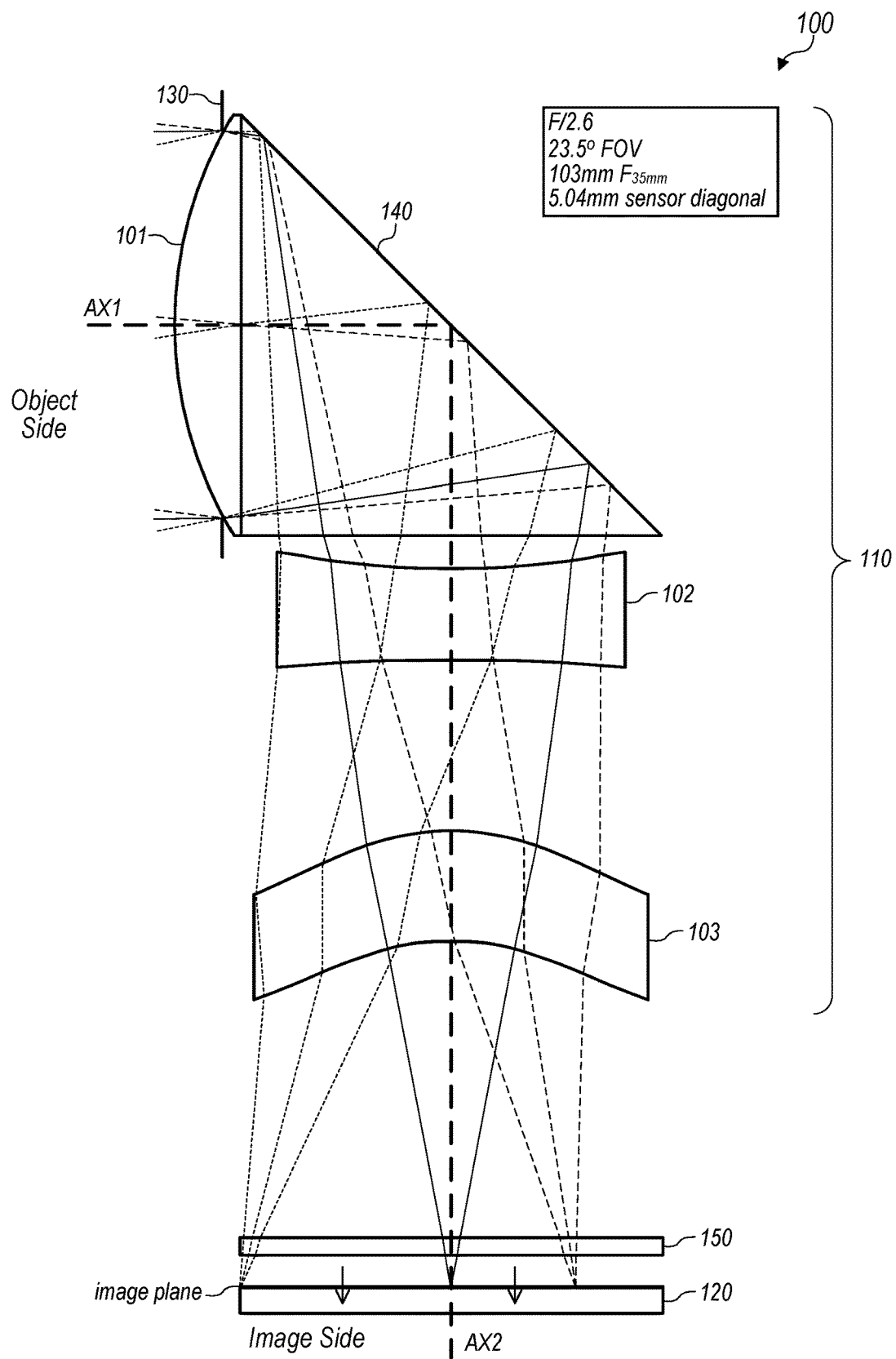


FIG. 1B

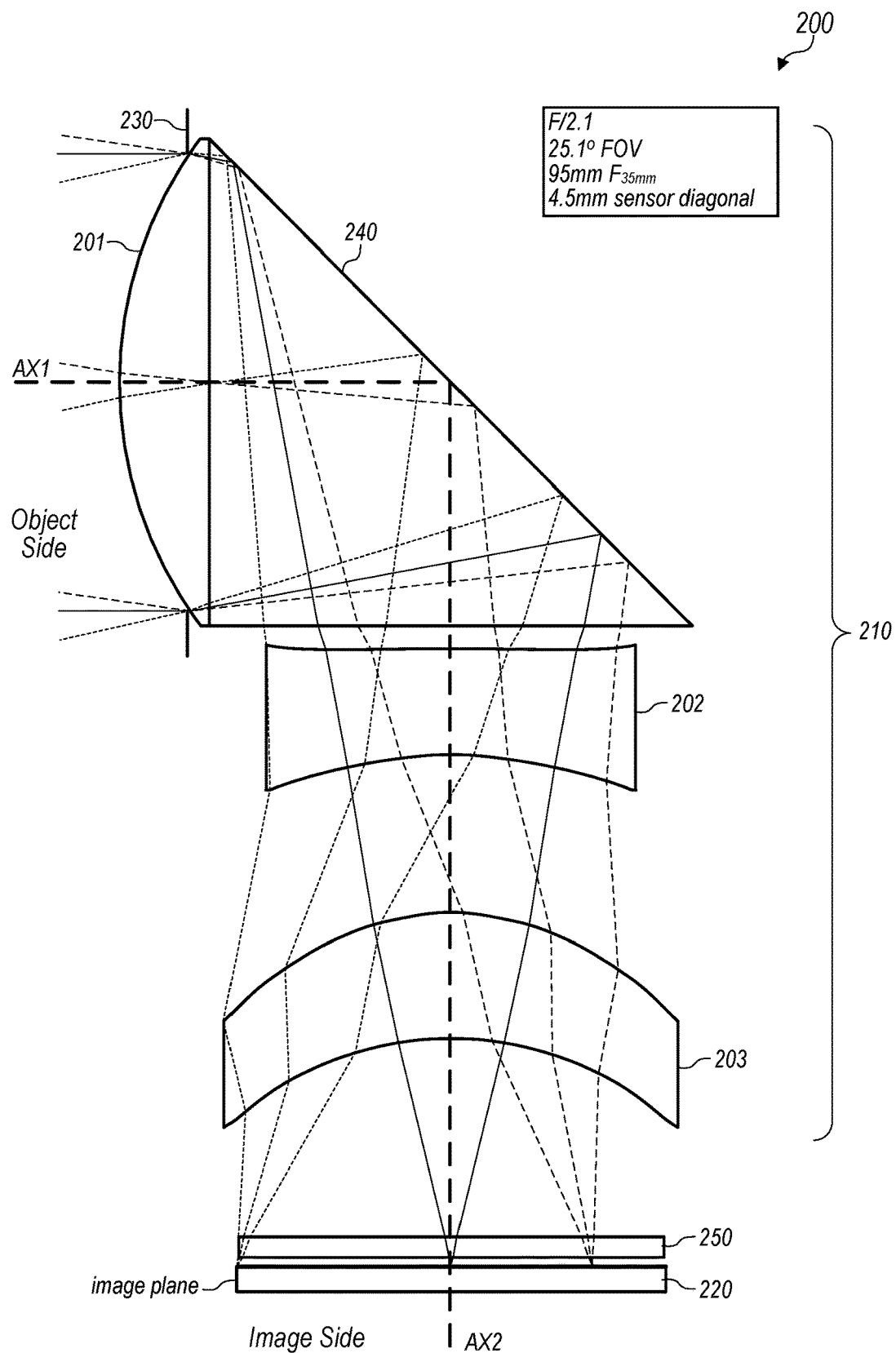


FIG. 2

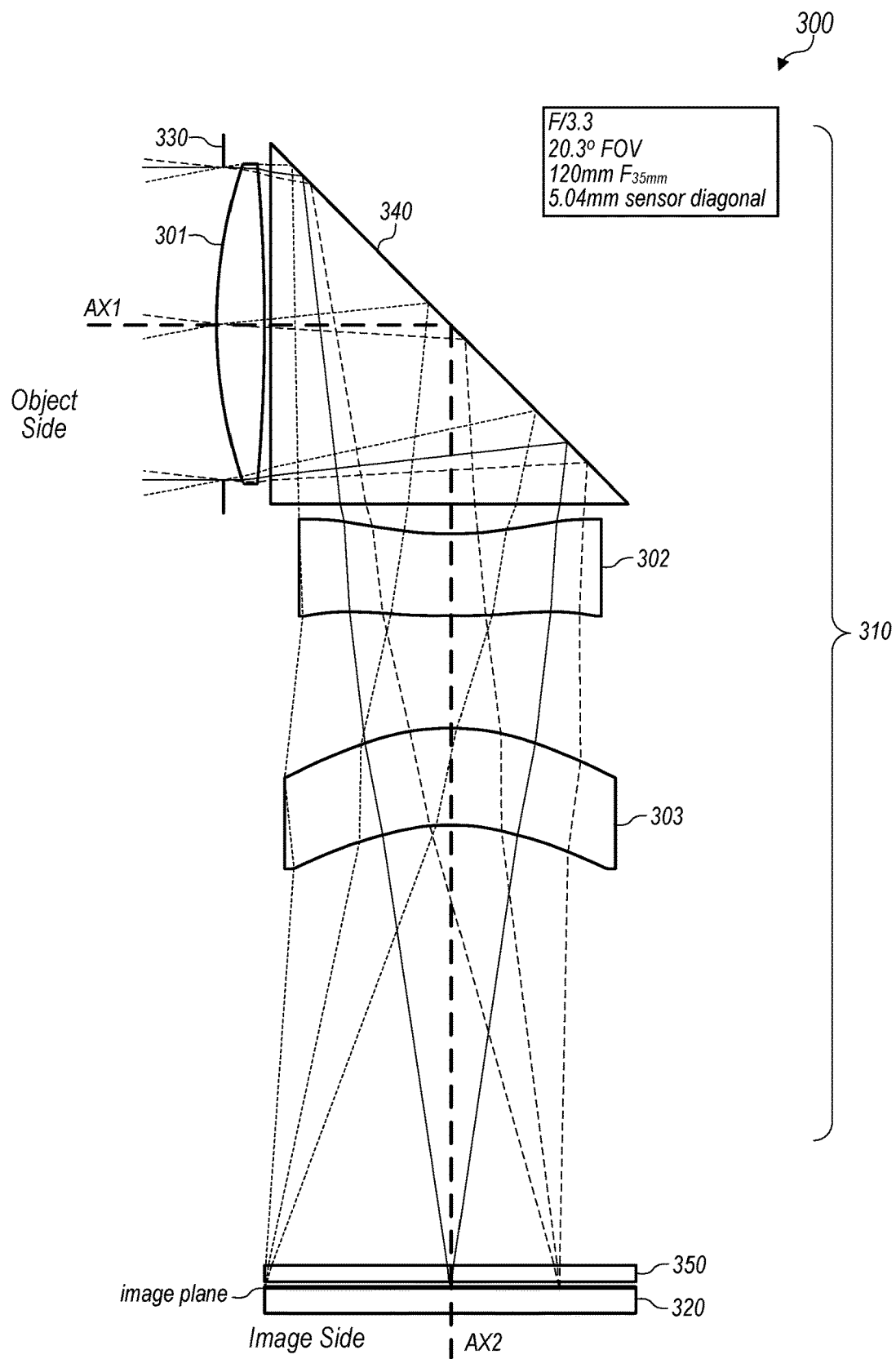


FIG. 3A

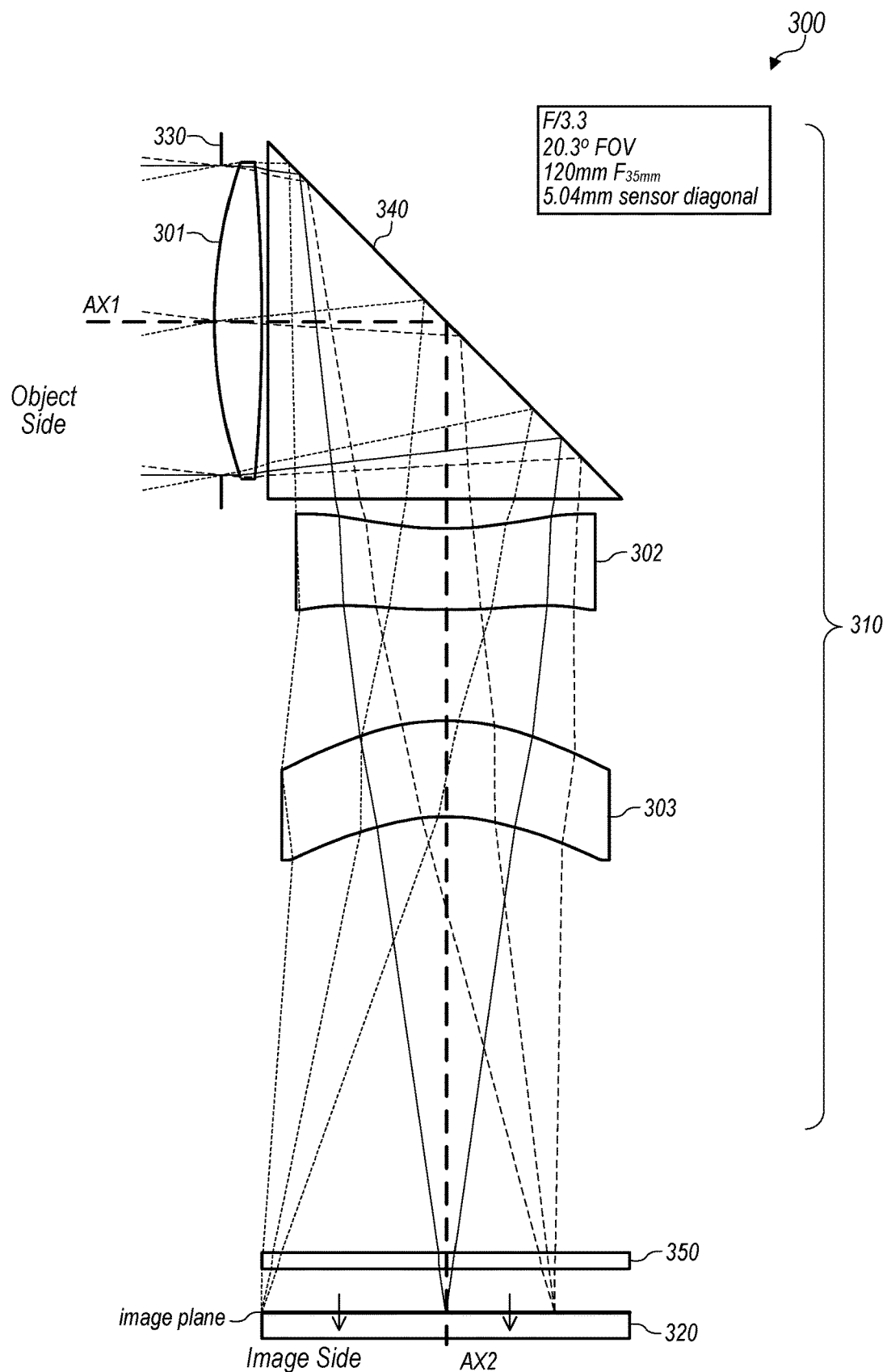


FIG. 3B

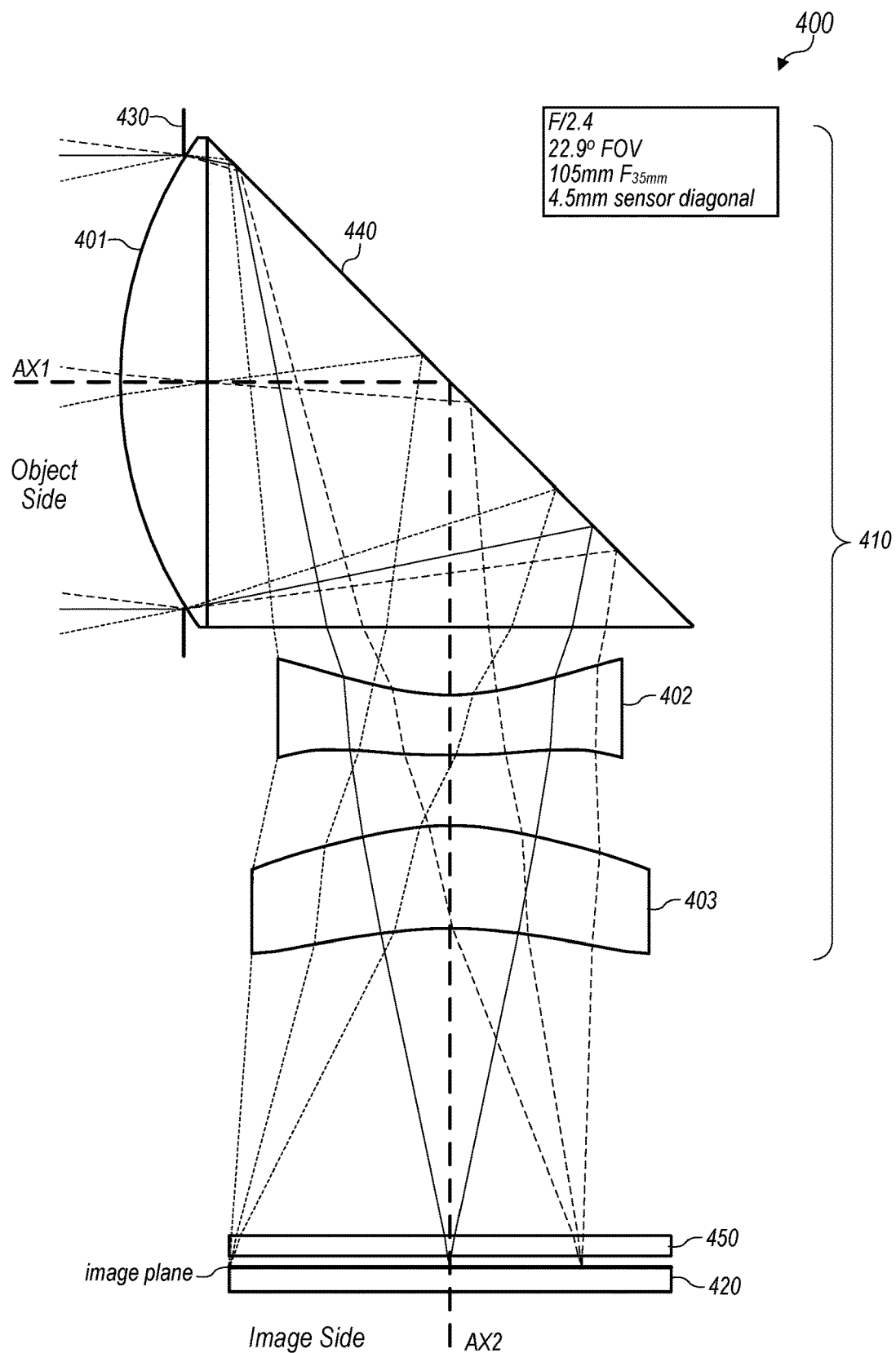


FIG. 4

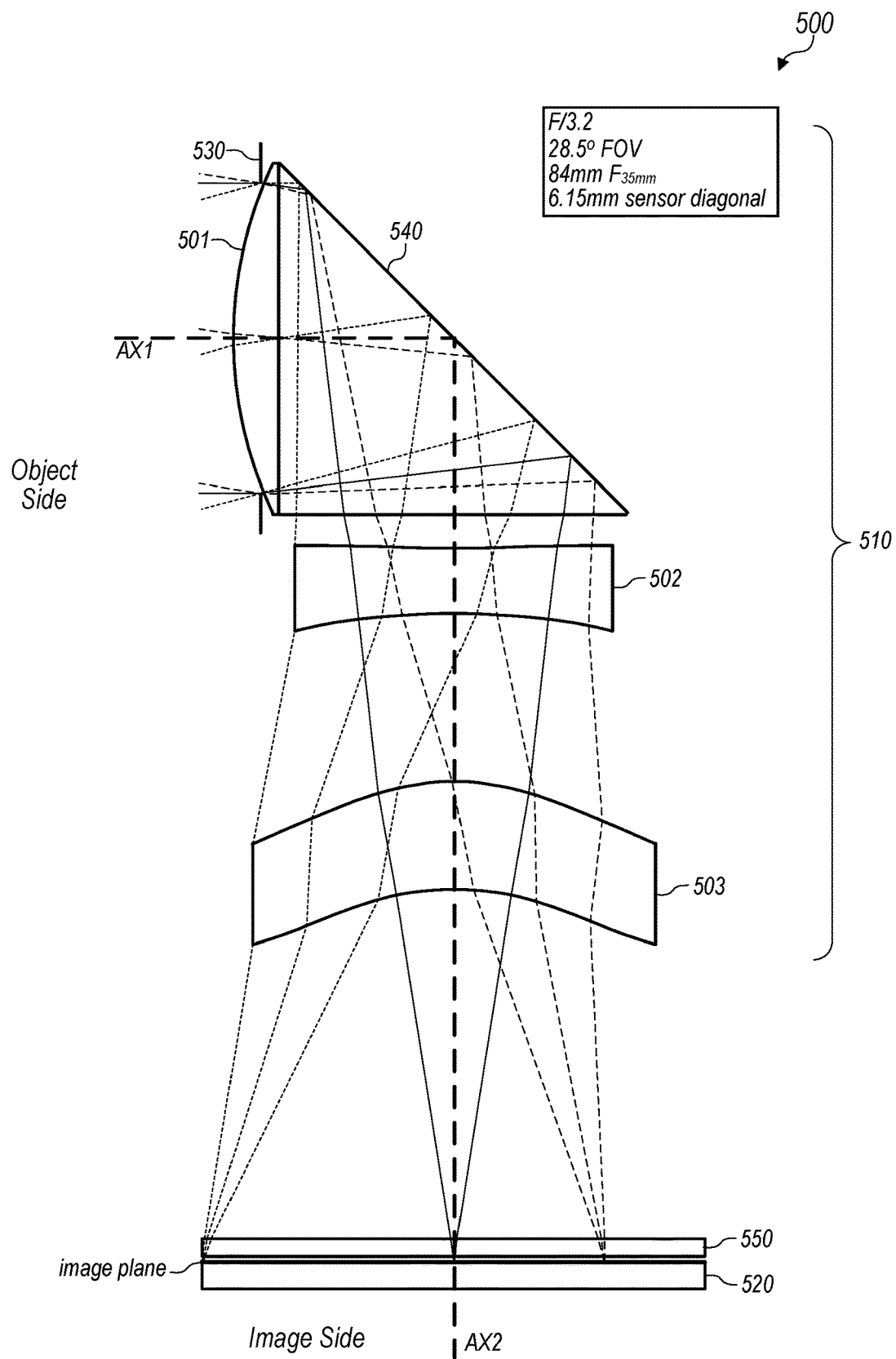


FIG. 5

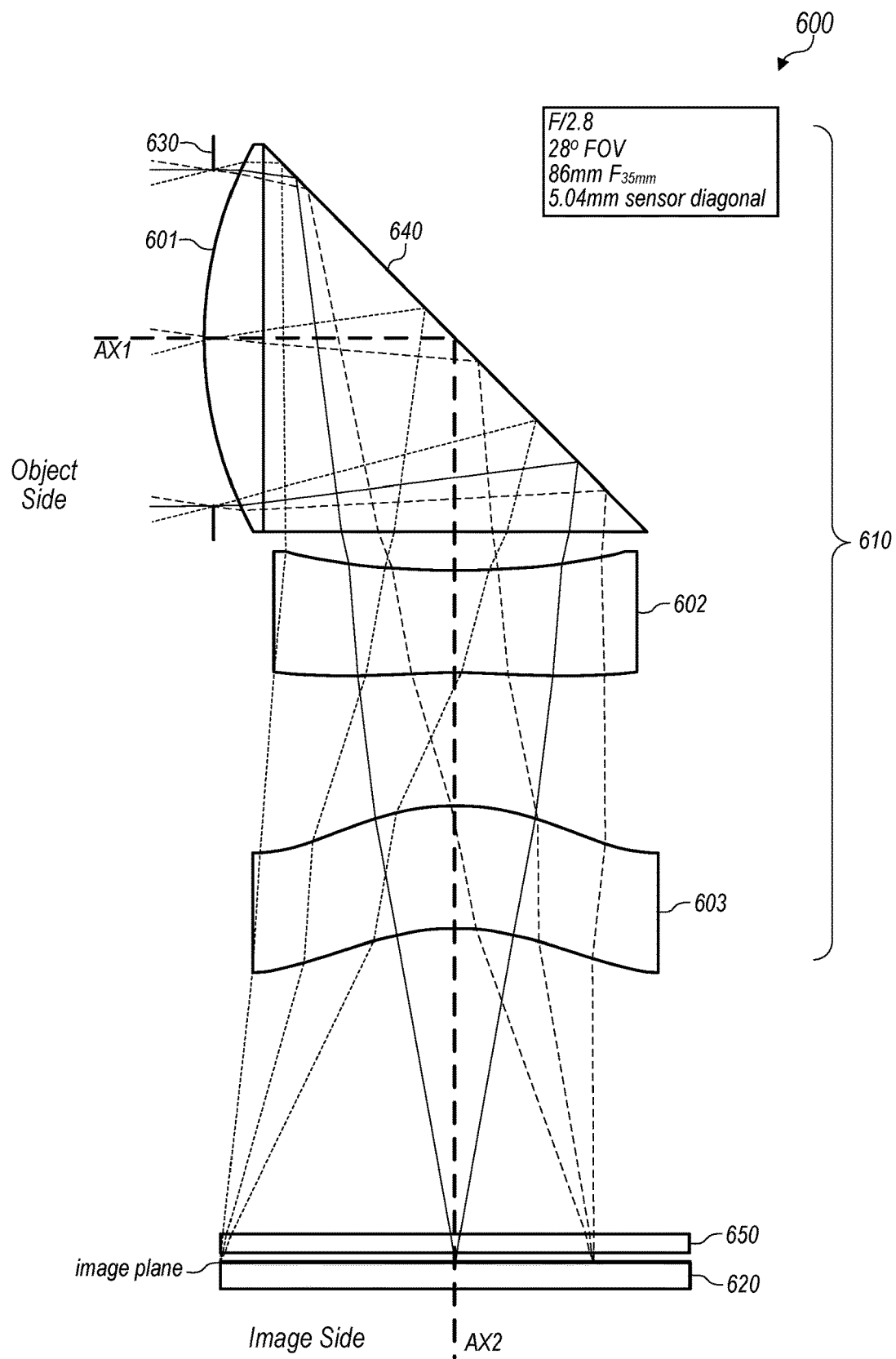


FIG. 6

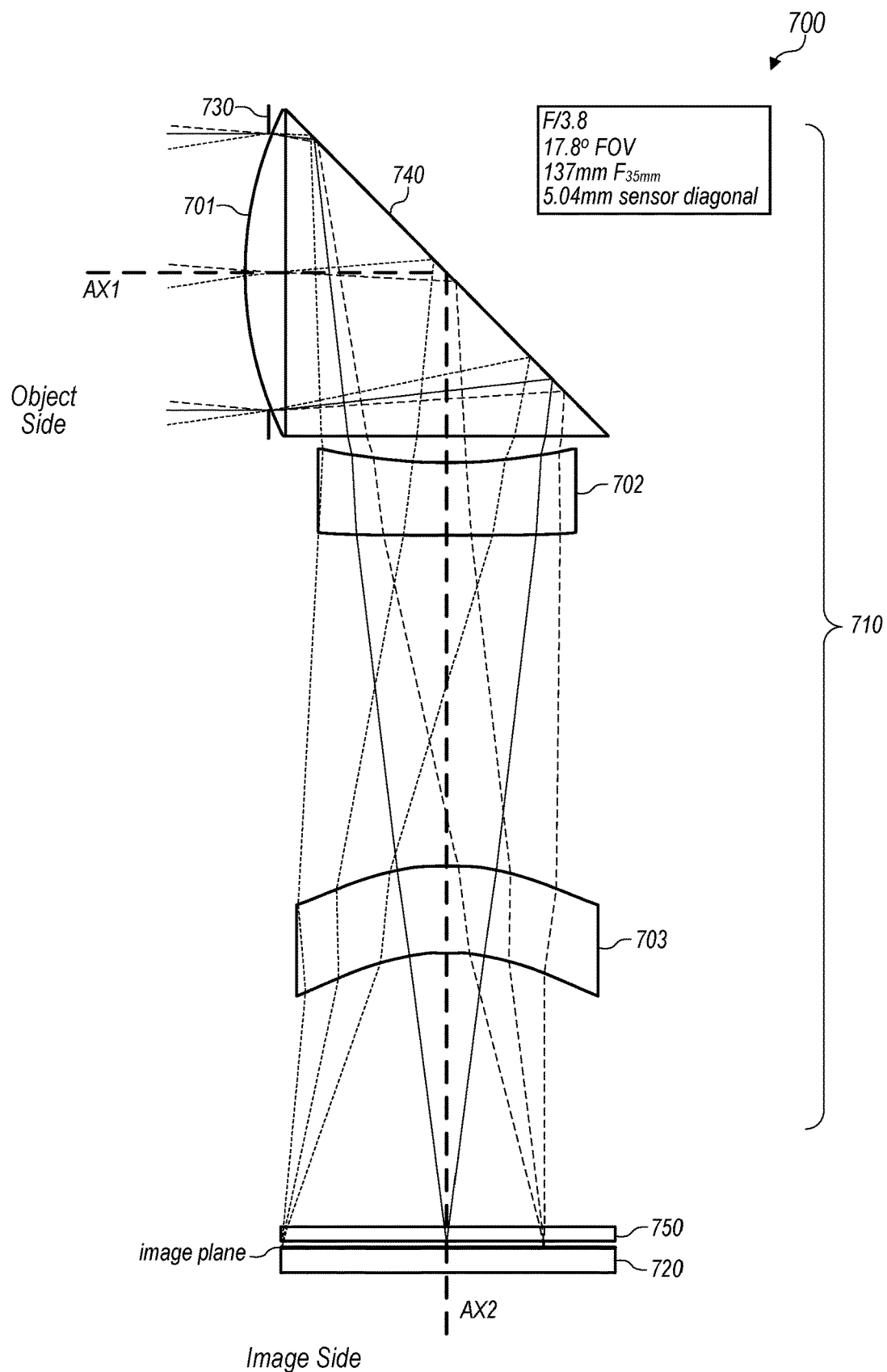


FIG. 7A

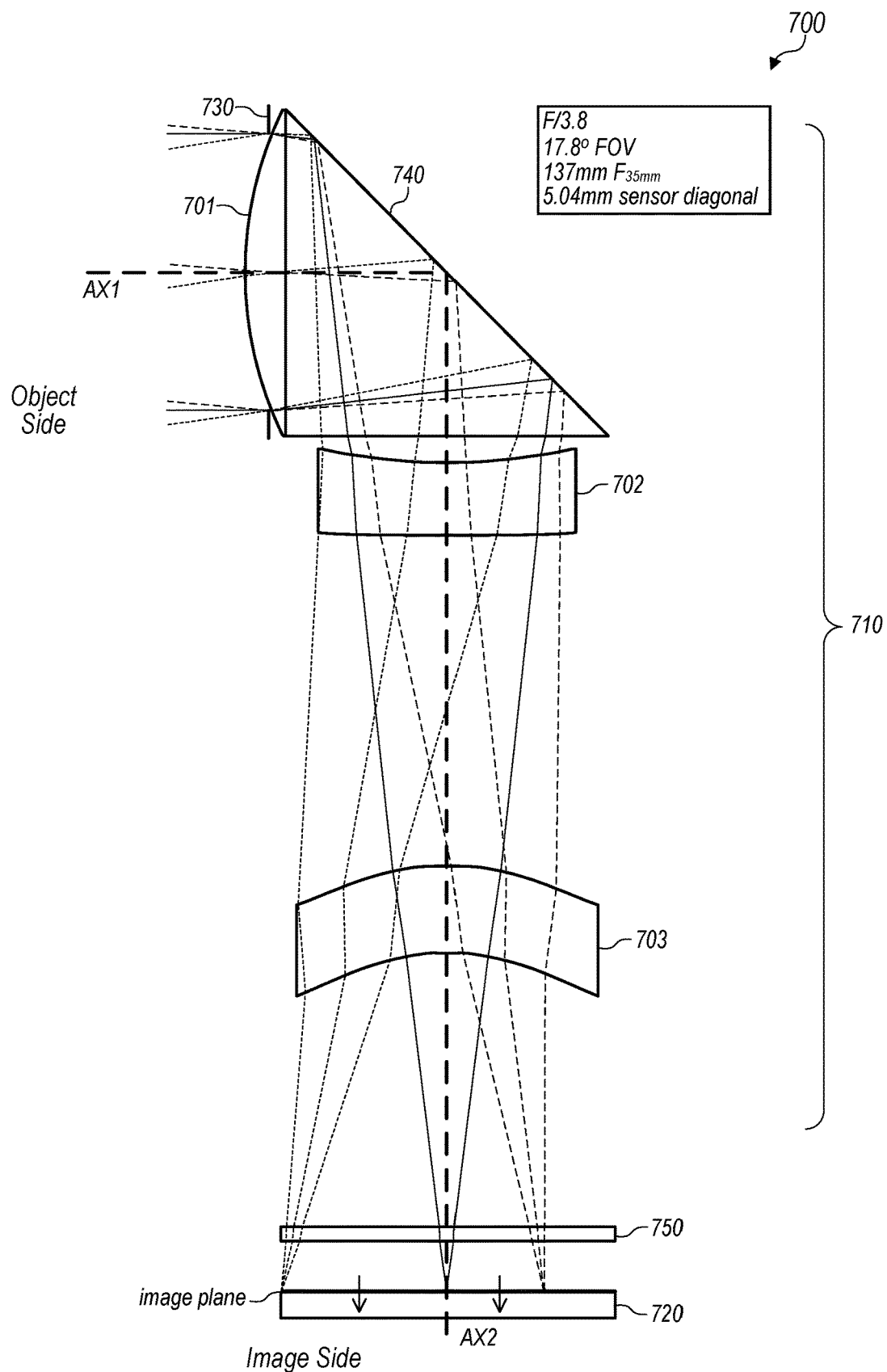


FIG. 7B

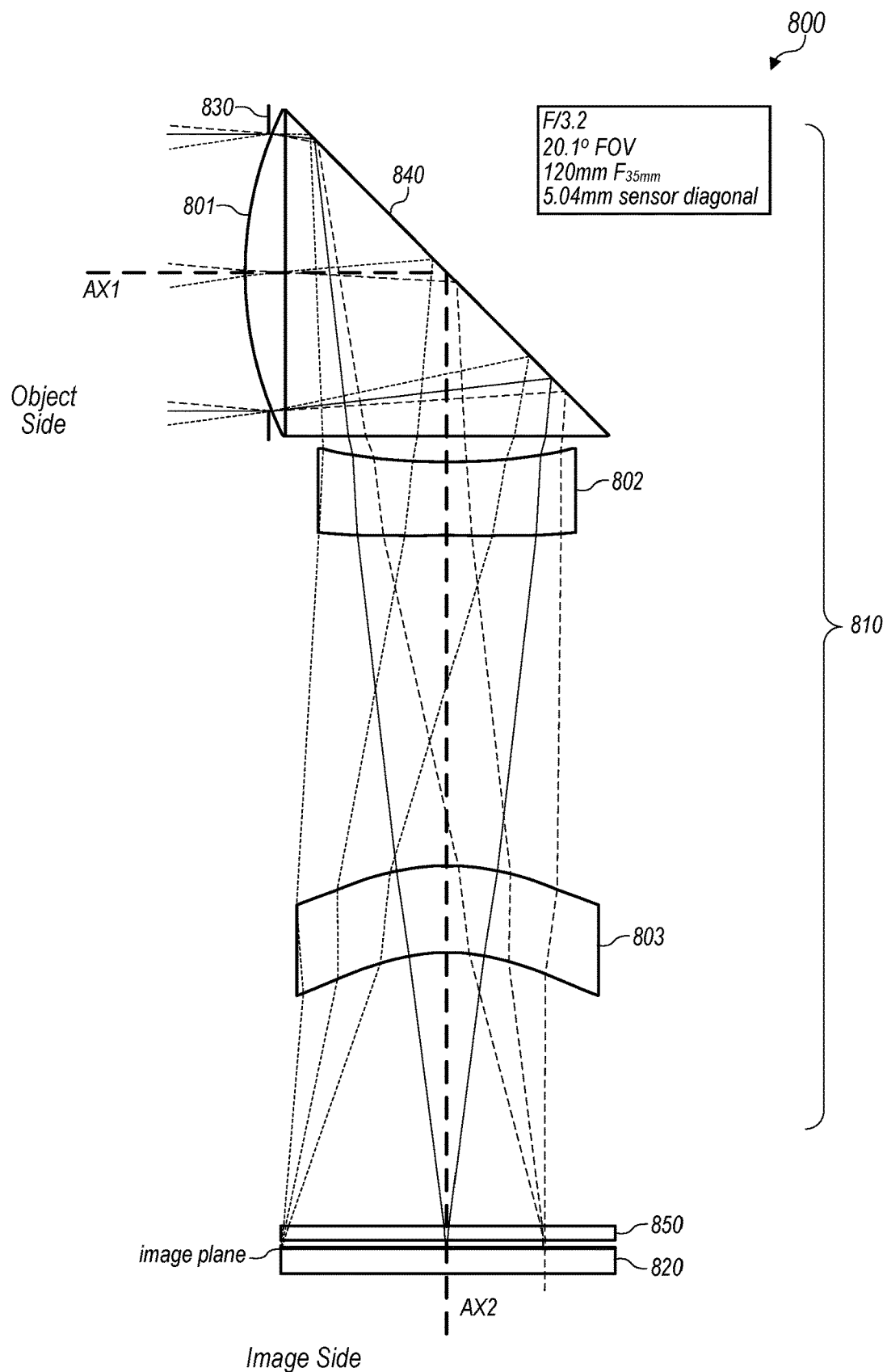


FIG. 8A

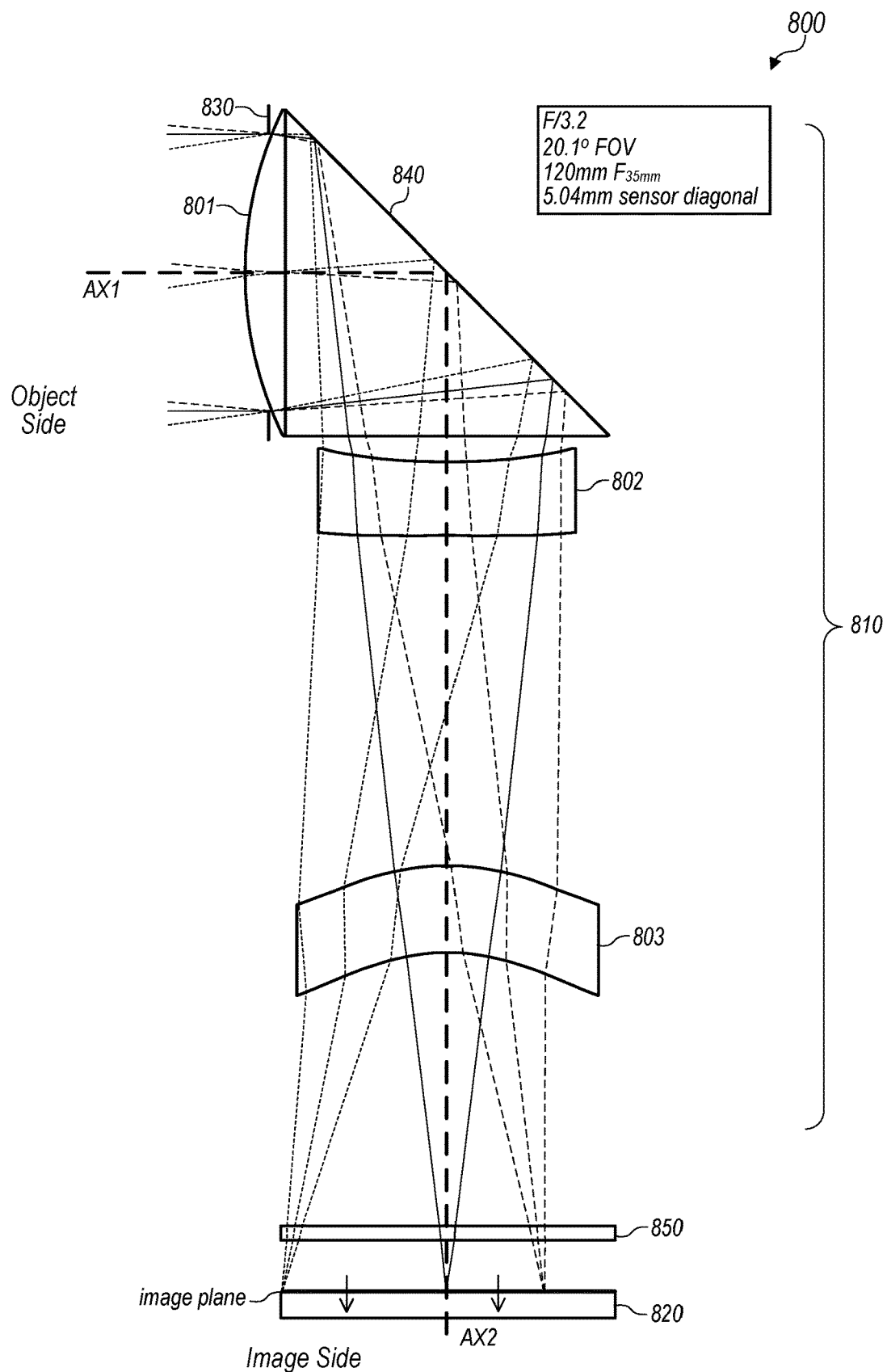


FIG. 8B

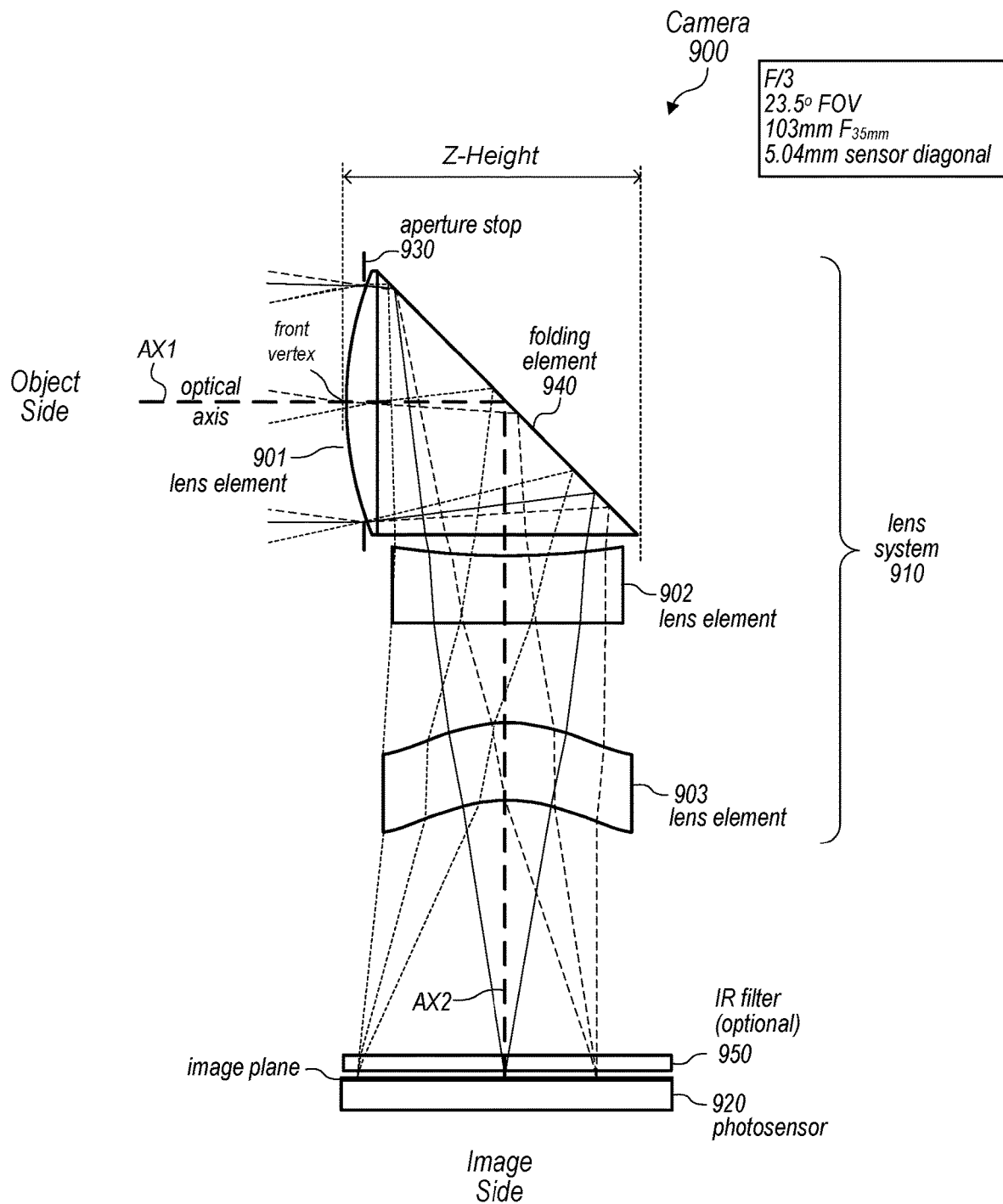


FIG. 9

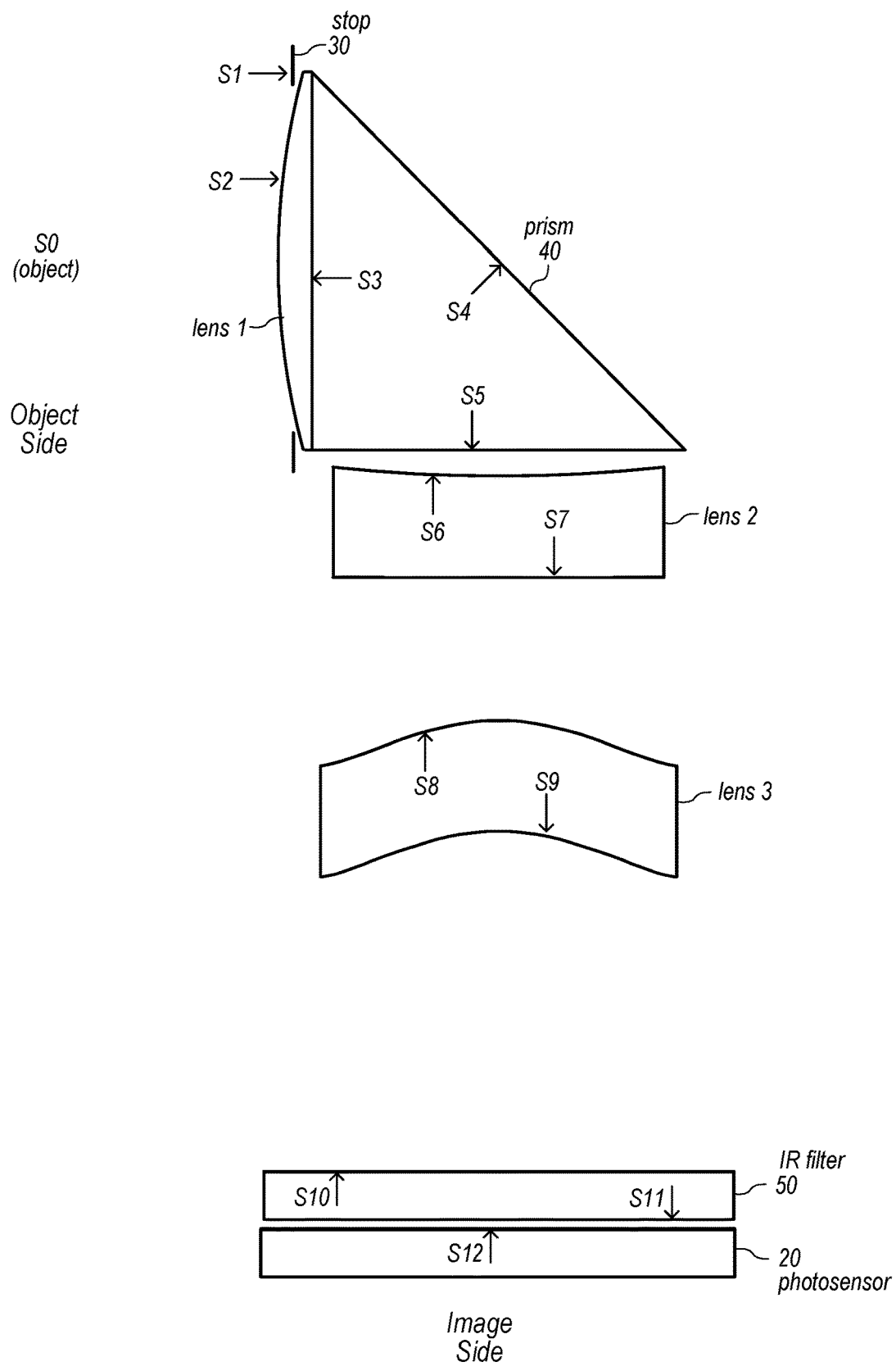


FIG. 10A

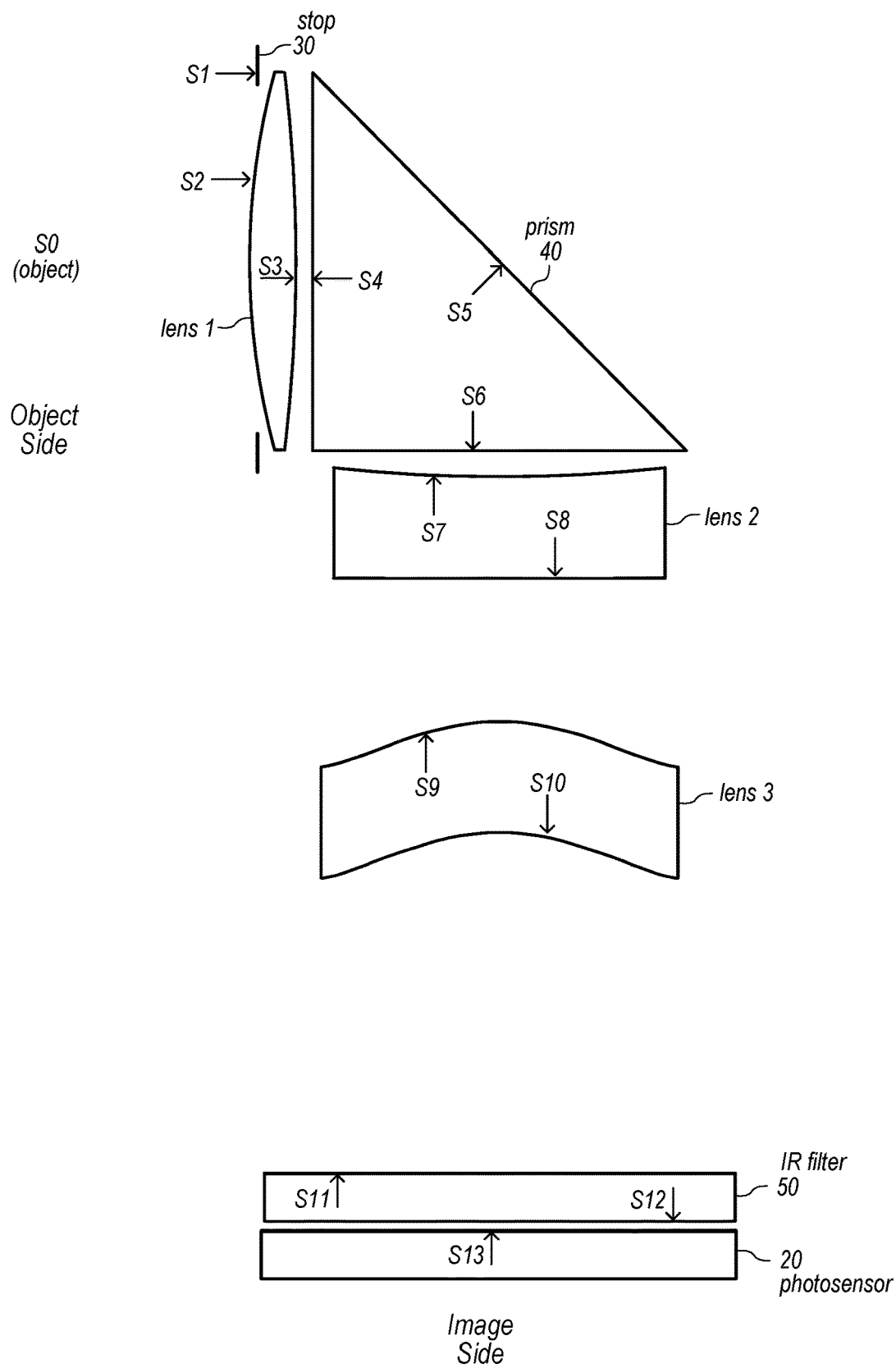


FIG. 10B

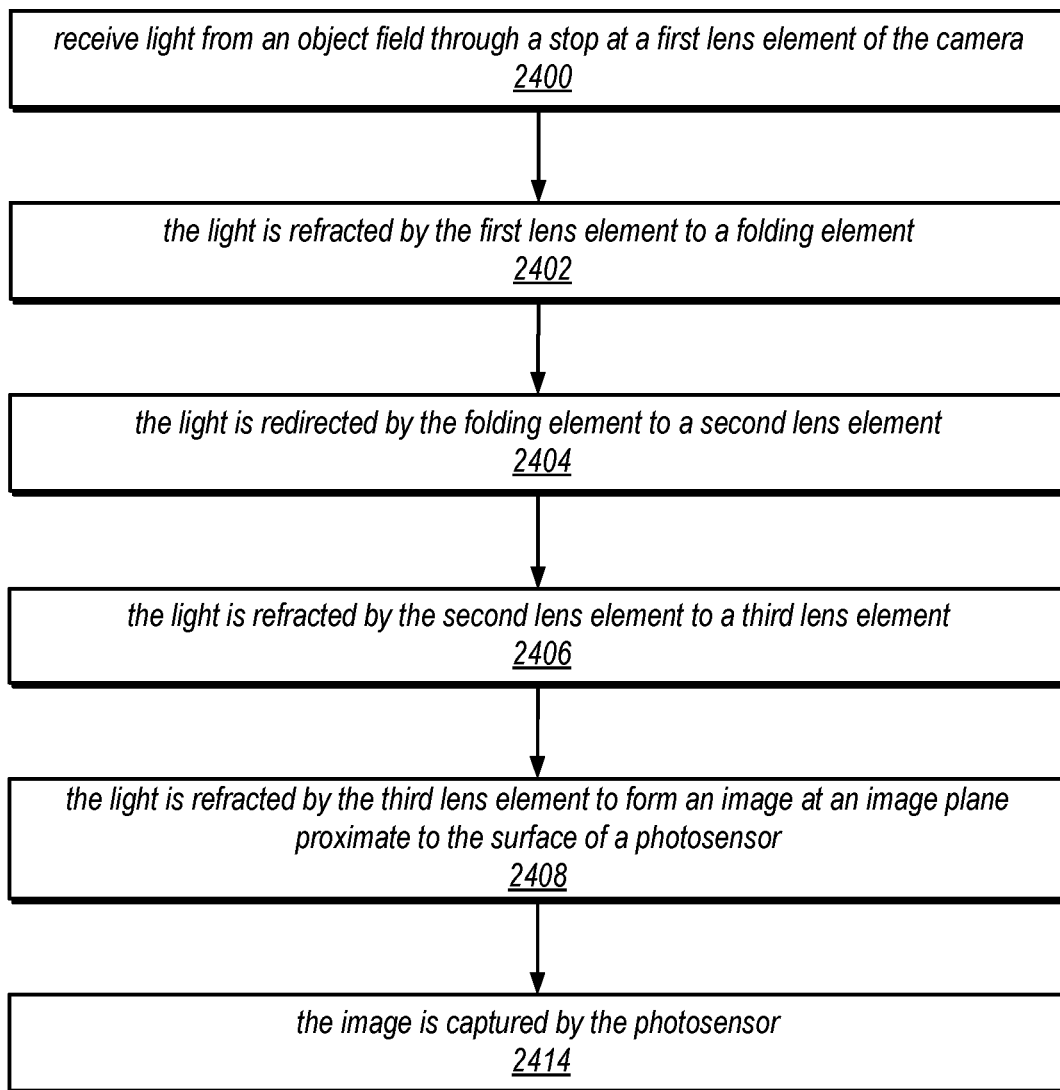


FIG. 11

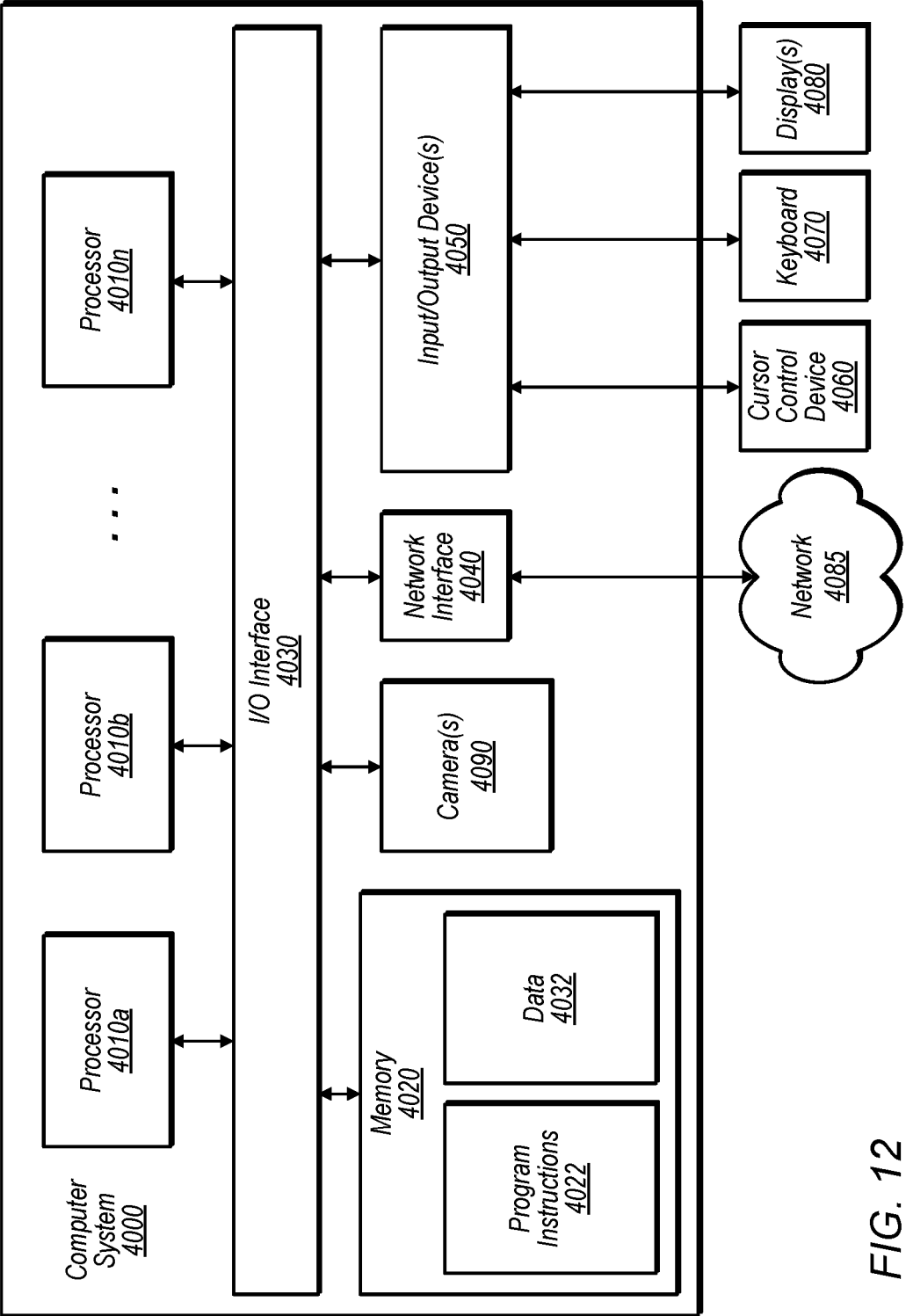


FIG. 12

FOLDED LENS SYSTEM WITH THREE REFRACTIVE LENSES

PRIORITY INFORMATION

[0001] This application is a continuation of U.S. patent application Ser. No. 16/593,888, filed Oct. 4, 2019, which is a continuation of U.S. patent application Ser. No. 15/472,138, filed Mar. 28, 2017, now U.S. Pat. No. 10,437,023, which claims benefit of priority of U.S. Provisional Application Ser. No. 62/314,350, filed Mar. 28, 2016, and also claims benefit of priority of U.S. Provisional Application Ser. No. 62/334,403, filed May 10, 2016, the content of which are incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

[0002] This disclosure relates generally to camera systems, and more specifically to compact lens systems for high-resolution, small form factor camera systems.

Description of the Related Art

[0003] The advent of small, mobile multipurpose devices such as smartphones and tablet or pad devices has resulted in a need for high-resolution, small form factor cameras for integration in the devices. However, due to limitations of conventional camera technology, conventional small cameras used in such devices tend to capture images at lower resolutions and/or with lower image quality than can be achieved with larger, higher quality cameras. Achieving higher resolution with small package size cameras generally requires use of a photosensor (also referred to as an image sensor) with small pixel size and a good, compact imaging lens system. Advances in technology have achieved reduction of the pixel size in photosensors. However, as photosensors become more compact and powerful, demand for compact imaging lens system with improved imaging quality performance has increased.

SUMMARY OF EMBODIMENTS

[0004] Compact folded lens systems are described that may be used in small form factor cameras. Lens systems are described that may include three lens elements with refractive power, with a light folding element such as a prism located between a first lens element on the object side of the lens system and a second lens element that redirects the light refracted from the first lens element from a first axis onto a second axis on which the other lens elements and a photosensor are arranged. The lens systems may include an aperture stop located behind the front vertex of the lens system, for example at the first lens element, and an optional infrared filter, for example located between the last lens element and a photosensor of the camera.

[0005] Embodiments of the compact folded lens system may include three lens elements with refractive power and a light folding element such as a prism to fold the optical axis. Embodiments of the compact folded lens system may be configured to operate with a relatively narrow field of view and a 35 mm equivalent focal length ($f_{35\text{ mm}}$) in the medium to long telephoto range. For example, some embodiments of the compact folded lens system may provide a 35 mm equivalent focal length in the range of 80-200

mm, with less than 6.5 mm of Z-height to fit in a wide variety of portable electronics devices.

[0006] Through proper arrangement in materials, power and radius of curvature of the three lens elements with power, embodiments of the compact folded lens are capable of capturing high resolution, high quality images at good brightness level. In some embodiments, a first lens element from the object side of the lens system has a convex object-side surface in the paraxial region, and a third lens element has a concave image-side surface in the paraxial region. In some embodiments, a first lens element from the object side of the lens system has a convex object-side surface in the paraxial region, and a third lens element has a concave image-side surface in the paraxial region and a convex object-side surface in the paraxial region (i.e., has a meniscus shape). In some embodiments, the first lens element is formed of an optical material with Abbe number $V_d > 40$, and a second lens element is formed of an optical material with Abbe number $V_d < 30$. In some embodiments, the first lens element is formed of an optical material with Abbe number $V_d > 45$, and a second lens element is formed of an optical material with Abbe number $V_d < 35$.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIGS. 1A and 1B is a cross-sectional illustration of a compact camera including an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/2.6, with 23.5° full field of view (FOV).

[0008] FIG. 2 shows a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/2.1, with 25.1° full FOV.

[0009] FIGS. 3A and 3B show a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/3.3, with 20.3° full FOV.

[0010] FIG. 4 shows a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/2.4, with 22.9° full FOV.

[0011] FIG. 5 shows a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/3.2, with 28.5° full FOV.

[0012] FIG. 6 shows a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/2.8, with 28° full FOV.

[0013] FIGS. 7A and 7B show a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/3.8, with 17.8° full FOV.

[0014] FIGS. 8A and 8B show a camera that includes an example embodiment of a compact folded lens system with three lens elements and a light folding element that operates at F/3.2, with 20.1° full FOV.

[0015] FIG. 9 is a cross-sectional illustration of a compact camera including an example embodiment of a compact folded lens system with three lens elements and a light folding element.

[0016] FIGS. 10A and 10B illustrate numbering of the surfaces in the example lens systems as used in the Tables.

[0017] FIG. 11 is a flowchart of a method for capturing images using cameras with lens systems as illustrated FIGS. 1 through 10B, according to some embodiments.

[0018] FIG. 12 illustrates an example computer system that may be used in embodiments.

[0019] This specification includes references to “one embodiment” or “an embodiment.” The appearances of the phrases “in one embodiment” or “in an embodiment” do not necessarily refer to the same embodiment. Particular features, structures, or characteristics may be combined in any suitable manner consistent with this disclosure.

[0020] “Comprising.” This term is open-ended. As used in the appended claims, this term does not foreclose additional structure or steps. Consider a claim that recites: “An apparatus comprising one or more processor units . . .”. Such a claim does not foreclose the apparatus from including additional components (e.g., a network interface unit, graphics circuitry, etc.).

[0021] “Configured To.” Various units, circuits, or other components may be described or claimed as “configured to” perform a task or tasks. In such contexts, “configured to” is used to connote structure by indicating that the units/circuits/components include structure (e.g., circuitry) that performs those task or tasks during operation. As such, the unit/circuit/component can be said to be configured to perform the task even when the specified unit/circuit/component is not currently operational (e.g., is not on). The units/circuits/components used with the “configured to” language include hardware—for example, circuits, memory storing program instructions executable to implement the operation, etc. Reciting that a unit/circuit/component is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112, sixth paragraph, for that unit/circuit/component. Additionally, “configured to” can include generic structure (e.g., generic circuitry) that is manipulated by software and/or firmware (e.g., an FPGA or a general-purpose processor executing software) to operate in manner that is capable of performing the task(s) at issue. “Configured to” may also include adapting a manufacturing process (e.g., a semiconductor fabrication facility) to fabricate devices (e.g., integrated circuits) that are adapted to implement or perform one or more tasks.

[0022] “First,” “Second,” etc. As used herein, these terms are used as labels for nouns that they precede, and do not necessarily imply any type of ordering (e.g., spatial, temporal, logical, etc.). For example, a buffer circuit may be described herein as performing write operations for “first” and “second” values. The terms “first” and “second” do not necessarily imply that the first value must be written before the second value.

[0023] “Based On.” As used herein, this term is used to describe one or more factors that affect a determination. This term does not foreclose additional factors that may affect a determination. That is, a determination may be solely based on those factors or based, at least in part, on those factors. Consider the phrase “determine A based on B.” While in this case, B is a factor that affects the determination of A, such a phrase does not foreclose the determination of A from also being based on C. In other instances, A may be determined based solely on B.

DETAILED DESCRIPTION

[0024] Embodiments of a compact folded lens system including three lens elements with refractive power, with a

light folding element such as a prism, located between a first lens element on the object side of the lens system and a second lens element, that redirects the light refracted from the first lens element from a first axis onto a second axis on which the other lens elements and a photosensor are arranged. The lens system may include an aperture stop, for example located at or behind the front vertex of the lens system, for example at the first lens element, and an optional infrared filter, for example located between the last lens element and the photosensor. The shapes, materials, and arrangements of the lens elements in the lens system may be selected to capture high resolution, high quality images.

[0025] Conventionally, compact imaging lenses can be designed with a non-folded optical axis that provide a 35 mm equivalent focal length ($f_{35\text{ mm}}$) of 50 mm-70 mm. However, the lens brightness (related to the focal ratio, or F/#, of the lens system) and image quality of these conventional compact lens designs are typically limited by the constraint in thickness (Z dimension) of portable electronics devices. It is difficult to further increase the lens effective focal length of these conventional compact lens designs due to the scaling relationship with respect to the lens dimensions. To overcome this limitation, a folding-prism or mirror may be used in embodiments to relieve the constraint in the Z dimension of the lens system.

[0026] Embodiments of the compact folded lens systems as described herein may provide high resolution, high quality imaging for small form factor cameras. Using an embodiment of the compact lens system, a camera may be implemented in a small package size while still capturing sharp, high-resolution images, making embodiments of the camera suitable for use in small and/or mobile multipurpose devices such as cell phones, smartphones, pad or tablet computing devices, laptop, netbook, notebook, subnotebook, and ultra-book computers, and so on. FIG. 12 illustrates an example device that may include one or more small form factor cameras that use embodiments of the compact folded lens systems as described herein. However, note that aspects of the camera (e.g., the lens system and photosensor) may be scaled up or down to provide cameras with larger or smaller package sizes. In addition, embodiments of the camera system may be implemented as stand-alone digital cameras. In addition to still (single frame capture) camera applications, embodiments of the camera system may be adapted for use in video camera applications.

Folded Lens Systems with Three Lens Elements

[0027] FIGS. 1A through 8B show several embodiments of compact cameras with compact folded lens systems with three lens elements and a light folding element such as a prism that “folds” the optical axis of the lens system. A compact camera including an embodiment of the compact folded lens systems as illustrated in FIGS. 1A through 8B may, for example, be implemented in portable electronic devices such as mobile phones and tablets. The lens system and/or camera may also include an aperture stop, an optional infrared (IR) filter, and a photosensor. The compact folded lens systems as illustrated in FIGS. 1A through 8B may be configured to operate with a relatively narrow field of view and a 35 mm equivalent focal length ($f_{35\text{ mm}}$) in the medium to long telephoto range. Compact cameras including the compact folded lens systems as illustrated in FIGS. 1A through 8B may, for example, be used stand-alone for telephoto photography, or can be paired with a wide-angle

imaging lens in a dual-prime configuration to enable effective optical zoom for portable electronic devices.

[0028] Embodiments of the compact folded lens system as illustrated in FIGS. 1A through 8B may include three lens elements with refractive power and a light folding element such as a prism to fold the optical axis. Embodiments of the compact folded lens system as illustrated in FIGS. 1A through 8B may provide a 35 mm equivalent focal length in the range of 80-200 mm and less than 6.5 mm of Z-height to fit in a wide variety of portable electronics devices. With proper arrangement in materials, power and radius of curvature of the three lens elements with power, embodiments of the compact folded lens system as illustrated in FIGS. 1A through 8B are capable of capturing high resolution, high quality images at good brightness level.

[0029] Embodiments of the compact folded lens system as illustrated in FIGS. 1A through 8B include three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element (lens 1) with positive refractive power, a folding element such as a prism to fold the optical axis from AX1 to AX2, a second lens element (lens 2) with negative refractive power, and a third lens element (lens 3) with refractive power. An aperture stop may be located between the object side of the lens system and the folding element for controlling the brightness of the optical system. In some embodiments, the lens system or camera includes an infrared (IR) filter to reduce or eliminate interference of environmental noises on the image sensor (also referred to herein as a photosensor or sensor). In some embodiments, the photosensor may be shifted along AX2 to allow refocusing of the lens system in between Infinity conjugate and Macro conjugate, for example for autofocus applications. Lens 2 and lens 3 may be round/circular optical lenses, or may have a shape other than circular (e.g., rectangular or square, hexagonal, etc.) to reduce the camera module Z height.

[0030] In embodiments of the compact folded lens system as illustrated in FIGS. 1A through 8B, one or more of the following requirements may be satisfied, for example to facilitate correction of aberrations across the field of view (FOV) for the lens system:

[0031] Lens 1 has a convex object-side surface in the paraxial region.

[0032] Lens 3 has a concave image-side surface in the paraxial region and a convex object-side surface in the paraxial region (i.e., lens 3 has a meniscus shape).

[0033] In various embodiments, the other lens surfaces of lenses 1 through 3 may be concave, convex, or flat/plano (e.g., the lenses may be plano-concave or plano-convex lenses) in the paraxial region.

[0034] In some embodiments, one or more of the following relationships may be met:

$$0.5 < |f/f_1| < 2$$

$$0.4 < |f/f_2| < 2.5$$

$$0.5 < |R_3/f| < 1.5$$

where f is effective focal length of the lens system, f_1 is focal length of lens 1, f_2 is focal length of lens 2, R_3 is radius of curvature of the object-side surface of lens 3, and R_{3r} is radius of curvature of the image side surface of lens 3.

[0035] In some embodiments, at least one of the six lens surfaces may be aspheric.

[0036] In some embodiments, at least one of the lens elements is made of lightweight polymer or plastic material.

[0037] In some embodiments, lens 1 is formed of an optical material with Abbe number $V_d > 45$, and lens 2 is formed of an optical material with Abbe number $V_d < 35$. The material and power configurations of lenses 1 and 2 may, for example, be selected for reduction of chromatic aberrations.

[0038] In some embodiments lens 3 is formed of an optical material with no limit in Abbe number.

[0039] As shown in the example embodiments in FIGS. 1A-1B, 3A-3B, 7A-7B, and 8A-8B, in some embodiments of a camera including compact folded lens system as illustrated in FIGS. 1A through 8B, the photosensor may be moved on one or more axes relative to the lens system to adjust focus of the camera. Alternatively, in some embodiments, the lens system may be moved relative to the photosensor to adjust focus. FIGS. 1A, 3A, 7A, and 8A correspond to the camera focused at a first position (infinity conjugate), and FIGS. 1B, 3B, 7B, and 8B correspond to the camera focused at a second position (e.g., macro conjugate, 500 mm in FIG. 1B). While the focus positions are shown as examples, note that the camera may be focused at other positions in some embodiments.

[0040] As shown in the example embodiments in FIGS. 1A-1B, 2, 4, 5, 6, 7A-7B, and 8A-8B, in some embodiments of a compact folded lens system as described herein, the image side surface of the first lens element (lens 1) may be flat/plano (e.g., lens 1 may be plano-convex), and the image side surface of lens 1 may be at/in contact with the object side surface of the light folding prism to effectively form a single combined unit or element. The lens 1 and prism elements may be composed of the same type of material (e.g., a plastic material) or of different types of materials. In some embodiments, the lens 1 and prism elements may be cemented. Alternatively, the lens 1 and prism elements may be composed of the same type of material (e.g., a plastic material), and may be molded as a single combined unit or element. However, in some embodiments, for example as shown in FIGS. 3A-3B, the image side surface of lens 1 may be convex, concave, or flat-plano, and lens 1 and the folding element (prism) may be air-spaced.

Example Lens System 110

[0041] FIGS. 1A and 1B show a camera 100 that includes an example embodiment of a compact folded lens system 110 that operates at F/2.6, with 23.5° full FOV. Camera 100 includes a 5.04 mm diagonal photosensor 120. Lens system 110 includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element 101 with positive refractive power, a folding element 140 such as a prism to fold the optical axis from AX1 to AX2, a second lens element 102 with negative refractive power, and a third lens element 103 with refractive power. An aperture stop 130 may be located between the object side of the lens system 110 and the folding element 140, for example at or near the object side surface of lens element 101, for controlling the brightness of the optical system. In some embodiments, the lens system 110 or camera 100

includes an IR filter **150** to reduce or eliminate interference of environmental noises on the photosensor **120**.

[0042] Tables 1-5 correspond to an embodiment of a lens system **110** as illustrated in FIGS. 1A and 1B, and provide example values for various optical and physical parameters of the lens system **110** and camera **100** of FIGS. 1A and 1B. The effective focal length (EFL) of the lens system **110** is 12 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **110** may be 103 mm. In some embodiments, the camera **100**/lens system **110** has the capability of autofocus from 500 mm to Infinity conjugates.

[0043] As shown in FIGS. 1A-1B, in some embodiments the photosensor **120** may be moved on one or more axes relative to the lens system **110** to adjust focus of the camera **100**. FIG. 1A corresponds to the camera **100** focused at a first position (infinity conjugate), and FIG. 1B corresponds to the camera **100** focused at a second position (500 mm in FIG. 1B). While the focus positions are shown as examples, note that the camera **100** may be focused at other positions in some embodiments.

[0044] The modulation transfer functions (MTFs) for lens system **110** when focused at Infinity and Macro (500 mm) conjugates, at all fields and both conjugates, are higher than 0.5 at 125 line pairs (lp)/mm and higher than 0.3 at 250 lp/mm; the lens system **110** provides good contrast for high-resolution imaging. At both conjugates, on-axis and off-axis aberrations for lens system **110** are well balanced across the FOV. At both conjugates, optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0045] In some embodiments, Z-height of the example lens system **110**, as defined from the front vertex of lens element **101** to the rear vertex of the folding element **140**, may be 5.8 mm. The lens system **110** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **210**

[0046] FIG. 2 shows a camera **200** that includes an example embodiment of a compact folded lens system **210** that operates at F/2.1, with 25.1° full FOV. Camera **200** includes a 4.5 mm diagonal photosensor **220**. Lens system **210** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element **201** with positive refractive power, a folding element **240** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **202** with negative refractive power, and a third lens element **203** with refractive power. An aperture stop **230** may be located between the object side of the lens system **210** and the folding element **240**, for example at or near the object side surface of lens element **201**, for controlling the brightness of the optical system. In some embodiments, the lens system **210** or camera **200** includes an IR filter **250** to reduce or eliminate interference of environmental noises on the photosensor **220**.

[0047] Tables 6-9 correspond to an embodiment of a lens system **210** as illustrated in FIG. 2, and provide example values for various optical and physical parameters of the lens system **210** and camera **200** of FIG. 2. The effective focal length (EFL) of the lens system **210** is 10 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **200** may be 95 mm. While not shown

in FIG. 2, in some embodiments, the camera **200**/lens system **210** has the capability of autofocus from Macro to Infinity conjugates.

[0048] The modulation transfer function (MTF) for lens system **210** is higher than 0.5 at 125 lp/mm and higher than 0.3 at 250 lp/mm; the lens system **210** provides good contrast for high-resolution imaging. On-axis and off-axis aberrations for lens system **210** are well balanced across the FOV. Optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0049] In some embodiments, Z-height of the example lens system **210**, as defined from the front vertex of lens element **201** to the rear vertex of the folding element **240**, may be 6 mm. The lens system **210** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **310**

[0050] FIGS. 3A and 3B show a camera **300** that includes an example embodiment of a compact folded lens system **310** that operates at F/3.3, with 20.3° full FOV. Camera **300** includes a 5.04 mm diagonal photosensor **320**. Lens system **310** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element **301** with positive refractive power, a folding element **340** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **302** with negative refractive power, and a third lens element **303** with refractive power. As shown in FIGS. 3A and 3B, the image side surface of lens element **301** is convex, and there is air space between lens element **301** and the object side surface of the folding element **340**. An aperture stop **330** may be located between the object side of the lens system **310** and the folding element **340**, for example at or near the front vertex of lens element **301**, for controlling the brightness of the optical system. In some embodiments, the lens system **310** or camera **300** includes an IR filter **350** to reduce or eliminate interference of environmental noises on the photosensor **320**.

[0051] Tables 10-14 correspond to an embodiment of a lens system **310** as illustrated in FIGS. 3A and 3B, and provide example values for various optical and physical parameters of the lens system **310** and camera **300** of FIGS. 3A and 3B. The effective focal length (EFL) of the lens system **310** is 14 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **310** may be 120 mm. In some embodiments, the camera **300**/lens system **310** has the capability of autofocus from Macro to Infinity conjugates.

[0052] As shown in FIGS. 3A-3B, in some embodiments the photosensor **320** may be moved on one or more axes relative to the lens system **310** to adjust focus of the camera **300**. FIG. 3A corresponds to the camera **300** focused at a first position (infinity conjugate), and FIG. 3B corresponds to the camera **300** focused at a second position (Macro conjugate). While the focus positions are shown as examples, note that the camera **300** may be focused at other positions in some embodiments.

[0053] The modulation transfer functions (MTFs) for lens system **310** when focused at Infinity and Macro conjugates, at all fields and both conjugates, are close to diffraction limited; the lens system **310** provides good contrast for

high-resolution imaging. At both conjugates, on-axis and off-axis aberrations for lens system **310** are well balanced across the FOV. At both conjugates, optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0054] In some embodiments, Z-height of the example lens system **310**, as defined from the front vertex of lens element **301** to the rear vertex of the folding element **340**, may be 5.6 mm. The lens system **310** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **410**

[0055] FIG. 4 shows a camera **400** that includes an example embodiment of a compact folded lens system **410** that operates at F/2.4, with 22.9° full FOV. Camera **400** includes a 4.5 mm diagonal photosensor **420**. Lens system **410** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element **401** with positive refractive power, a folding element **440** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **402** with negative refractive power, and a third lens element **403** with refractive power. An aperture stop **430** may be located between the object side of the lens system **410** and the folding element **440**, for example at or near the object side surface of lens element **401**, for controlling the brightness of the optical system. In some embodiments, the lens system **410** or camera **400** includes an IR filter **450** to reduce or eliminate interference of environmental noises on the photosensor **420**.

[0056] Tables 15-18 correspond to an embodiment of a lens system **410** as illustrated in FIG. 4, and provide example values for various optical and physical parameters of the lens system **410** and camera **400** of FIG. 4. The effective focal length (EFL) of the lens system **410** is 11 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **410** may be 105 mm. While not shown in FIG. 4, in some embodiments, the camera **400**/lens system **410** has the capability of autofocus from Macro to Infinity conjugates.

[0057] On-axis and off-axis aberrations for lens system **410** are well balanced across the FOV. Optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0058] In some embodiments, Z-height of the example lens system **410**, as defined from the front vertex of lens element **401** to the rear vertex of the folding element **440**, may be 5.85 mm. The lens system **410** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **510**

[0059] FIG. 5 shows a camera **500** that includes an example embodiment of a compact folded lens system **510** that operates at F/3.2, with 28.5° full FOV. Camera **500** includes a 6.15 mm diagonal photosensor **520**. Lens system **510** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element **501** with positive refractive power, a folding element **540** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **502** with negative refractive power,

and a third lens element **503** with refractive power. An aperture stop **530** may be located between the object side of the lens system **510** and the folding element **540**, for example at or near the object side surface of lens element **501**, for controlling the brightness of the optical system. In some embodiments, the lens system **510** or camera **500** includes an IR filter **550** to reduce or eliminate interference of environmental noises on the photosensor **520**.

[0060] Tables 19-22 correspond to an embodiment of a lens system **510** as illustrated in FIG. 5, and provide example values for various optical and physical parameters of the lens system **510** and camera **500** of FIG. 5. The effective focal length (EFL) of the lens system **510** is 12 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **510** may be 84 mm. While not shown in FIG. 5, in some embodiments, the camera **500**/lens system **510** has the capability of autofocus from Macro to Infinity conjugates.

[0061] On-axis and off-axis aberrations for lens system **510** are well balanced across the FOV. Optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0062] In some embodiments, Z-height of the example lens system **510**, as defined from the front vertex of lens element **501** to the rear vertex of the folding element **540**, may be 4.85 mm. The lens system **510** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **610**

[0063] FIG. 6 shows a camera **600** that includes an example embodiment of a compact folded lens system **610** that operates at F/2.8, with 28° full FOV. Camera **600** includes a 5.04 mm diagonal photosensor **620**. Lens system **610** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element **601** with positive refractive power, a folding element **640** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **602** with negative refractive power, and a third lens element **603** with refractive power. An aperture stop **630** may be located between the object side of the lens system **610** and the folding element **640**, for example at or near the front vertex of lens element **601**, for controlling the brightness of the optical system. In some embodiments, the lens system **610** or camera **600** includes an IR filter **650** to reduce or eliminate interference of environmental noises on the photosensor **620**.

[0064] Tables 23-26 correspond to an embodiment of a lens system **610** as illustrated in FIG. 6, and provide example values for various optical and physical parameters of the lens system **610** and camera **600** of FIG. 6. The effective focal length (EFL) of the lens system **610** is 10 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **610** may be 86 mm. While not shown in FIG. 6, in some embodiments, the camera **600**/lens system **610** has the capability of autofocus from Macro to Infinity conjugates.

[0065] On-axis and off-axis aberrations for lens system **610** are well balanced across the FOV. Optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0066] In some embodiments, Z-height of the example lens system **610**, as defined from the front vertex of lens

element **601** to the rear vertex of the folding element **640**, may be 4.75 mm. The lens system **610** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **710**

[0067] FIGS. 7A and 7B show a camera **700** that includes an example embodiment of a compact folded lens system **710** that operates at F/3.8, with 17.8° full FOV. Camera **700** includes a 5.04 mm diagonal photosensor **720**. Lens system **710** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element **701** with positive refractive power, a folding element **740** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **702** with negative refractive power, and a third lens element **703** with refractive power. An aperture stop **730** may be located between the object side of the lens system **710** and the folding element **740**, for example at or near the object side surface of lens element **701**, for controlling the brightness of the optical system. In some embodiments, the lens system **710** or camera **700** includes an IR filter **750** to reduce or eliminate interference of environmental noises on the photosensor **720**.

[0068] Tables 27-30 correspond to an embodiment of a lens system **710** as illustrated in FIGS. 7A and 7B, and provide example values for various optical and physical parameters of the lens system **710** and camera **700** of FIGS. 7A and 7B. The effective focal length (EFL) of the lens system **710** is 16 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **710** may be 137 mm. In some embodiments, the camera **700**/lens system **710** has the capability of autofocus from Macro to Infinity conjugates.

[0069] As shown in FIGS. 7A-7B, in some embodiments the photosensor **720** may be moved on one or more axes relative to the lens system **710** to adjust focus of the camera **700**. FIG. 7A corresponds to the camera **300** focused at a first position (infinity conjugate), and FIG. 7B corresponds to the camera **700** focused at a second position (Macro conjugate). While the focus positions are shown as examples, note that the camera **700** may be focused at other positions in some embodiments.

[0070] At both Infinity and Macro conjugates, on-axis and off-axis aberrations for lens system **710** are well balanced across the FOV. At both conjugates, optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0071] In some embodiments, Z-height of the example lens system **710**, as defined from the front vertex of lens element **701** to the rear vertex of the folding element **740**, may be 5.5 mm. The lens system **710** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System **810**

[0072] FIGS. 8A and 8B show a camera **800** that includes an example embodiment of a compact folded lens system **810** that operates at F/3.2, with 20.1° full FOV. Camera **800** includes a 5.04 mm diagonal photosensor **820**. Lens system **810** includes three lens elements with refractive power and a folding element such as a prism, in order from the object side to the image side of the lens system: a first lens element

801 with positive refractive power, a folding element **840** such as a prism to fold the optical axis from AX1 to AX2, a second lens element **802** with negative refractive power, and a third lens element **803** with refractive power. An aperture stop **830** may be located between the object side of the lens system **810** and the folding element **840**, for example at or near the object side surface of lens element **801**, for controlling the brightness of the optical system. In some embodiments, the lens system **810** or camera **800** includes an IR filter **850** to reduce or eliminate interference of environmental noises on the photosensor **820**.

[0073] Tables 31-34 correspond to an embodiment of a lens system **810** as illustrated in FIGS. 8A and 8B, and provide example values for various optical and physical parameters of the lens system **810** and camera **800** of FIGS. 8A and 8B. The effective focal length (EFL) of the lens system **810** is 14 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **810** may be 120 mm. In some embodiments, the camera **800**/lens system **810** has the capability of autofocus from Macro to Infinity conjugates.

[0074] As shown in FIGS. 8A-8B, in some embodiments the photosensor **820** may be moved on one or more axes relative to the lens system **810** to adjust focus of the camera **800**. FIG. 8A corresponds to the camera **800** focused at a first position (infinity conjugate), and FIG. 8B corresponds to the camera **800** focused at a second position (Macro conjugate). While the focus positions are shown as examples, note that the camera **700** may be focused at other positions in some embodiments.

[0075] At both Infinity and Macro conjugates, on-axis and off-axis aberrations for lens system **810** are well balanced across the FOV. At both conjugates, optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0076] In some embodiments, Z-height of the example lens system **810**, as defined from the front vertex of lens element **801** to the rear vertex of the folding element **840**, may be 5.4 mm. The lens system **810** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Folded Lens Systems with Three Lens Elements—Alternative Embodiments

[0077] FIG. 9 is a cross-sectional illustration of a compact camera **900** including an example embodiment of a compact folded lens system **910** with three lens elements **901-903** and a light folding element **940** such as a prism that “folds” the optical axis of the lens system **910**. The camera **900** may also include an aperture stop **930**, an optional infrared (IR) filter **950**, and a photosensor **920**. A compact camera **900** including an embodiment of the compact folded lens system **910** as illustrated in FIG. 9 may, for example, be implemented in portable electronic devices such as mobile phones and tablets. In embodiments of a lens system **910** as illustrated in FIG. 9, the 35 millimeter (mm) equivalent focal length ($f_{35\text{ mm}}$) of the lens system **910** may be longer than 60 mm. A compact folded lens system **910** having a long $f_{35\text{ mm}}$ may, for example, be used stand-alone for telephoto photography, or can be paired with a wide-angle imaging lens in a dual-prime configuration to enable effective optical zoom for portable electronic devices.

[0078] Embodiments of the compact folded lens system **910** may include three lens elements **901-903** with refractive power and a light folding element **940** such as a prism to fold

the optical axis. Some embodiments of the compact folded lens system **910** may provide a 35 mm equivalent focal length in the range of 85-200 mm and less than 6 mm of Z-height to fit in a wide variety of portable electronics devices. With proper arrangement of materials and lens powers, embodiments of the compact folded lens system **910** are capable of capturing high brightness photos with near diffraction-limited image quality.

[0079] As illustrated in the example camera **900** of FIG. 9, the compact folded lens system **910** includes three lens elements **901-903** with refractive power and a light folding element **940** (e.g., a prism), in order from the object side to the image side of the lens system **910**: a first lens element **901** with positive refractive power; a folding element **940** such as a prism to fold the optical axis from AX1 to AX2; a second lens element **902** with negative refractive power; and a third lens element **903** with refractive power. An aperture stop **930** may be located between the object side of the lens system **910** and the folding element **940** for controlling the brightness of the lens system **910**.

[0080] In some embodiments, the camera **900** includes an IR filter **950** to reduce or eliminate interference of environmental noises on the photosensor **920**. In some embodiments, the photosensor **920** and/or lens system **910** may be shifted along AX2 to allow refocusing of the lens system **910** in between Infinity conjugate and Macro conjugate. In various embodiments, lens element **902** and/or lens element **903** may be round/circular or rectangular, or some other shape.

[0081] In embodiments of lens system **910**, one or more of the following requirements may be satisfied, for example to facilitate correction of aberrations across the field of view (FOV) for the lens system **910**:

[0082] Lens element **901** has a convex object-side surface in the paraxial region.

[0083] Lens element **903** has a concave image-side surface in the paraxial region.

[0084] In various embodiments, the other lens surfaces of lens elements **901** through **903** may be concave, convex, or flat/plano (e.g., the lenses may be plano-concave or plano-convex lenses) in the paraxial region.

[0085] In some embodiments, at least one of the six lens surfaces may be aspheric.

[0086] In some embodiments, at least one of the lens elements is made of a lightweight polymer or plastic material.

[0087] In some embodiments, lens element **901** is formed of an optical material with Abbe number $V_d > 40$, and lens element **902** is formed of an optical material with Abbe number $V_d < 30$. The material and power configurations of lens element **901** and lens element **902** are selected for reduction of chromatic aberrations.

[0088] In some embodiments, lens element **903** is formed of an optical material with no limit in Abbe number.

[0089] FIG. 9 shows an example camera **900** that includes an example embodiment of a compact folded lens system **910** that operates at F/3, with 23.5° full FOV. Camera **900** includes a 5.04 mm diagonal photosensor **920**. The effective focal length (EFL) of the lens system **910** is 12 mm. Given the EFL and photosensor size, the 35 mm equivalent focal length of the camera **910** may be as large as 103 mm. In

some embodiments, the camera **900**/lens system **910** has the capability of autofocus from 300 mm to Infinity conjugates.

[0090] The modulation transfer functions (MTFs) for lens system **910** when focused at Infinity and Macro (300 mm) conjugates, at all fields and both conjugates, are close to diffraction limited; the lens system **910** provides good contrast for high-resolution imaging. At both conjugates, on-axis and off-axis aberrations for lens system **910** are well balanced across the FOV. At both conjugates, optical distortion across the FOV is controlled within 2%, while field curvature and astigmatism are well balanced across the FOV.

[0091] In some embodiments, Z-height of the example lens system **910**, as defined from the front vertex of lens element **901** to the rear vertex of the folding element **940**, may be 5 mm. The lens system **910** is able to fit into a wide variety of portable electronic devices including but not limited to smart phones and tablets.

Example Lens System Tables

[0092] The following Tables provide example values for various optical and physical parameters of the example embodiments of the lens systems and cameras as described in reference to FIGS. 1A through 9. Tables 1-5 correspond to an example embodiment of a lens system **110** as illustrated in FIGS. 1A and 1B. Tables 6-9 correspond to an example embodiment of a lens system **210** as illustrated in FIG. 2. Tables 10-14 correspond to an example embodiment of a lens system **310** as illustrated in FIGS. 3A and 3B. Tables 15-18 correspond to an example embodiment of a lens system **410** as illustrated in FIG. 4. Tables 19-22 correspond to an example embodiment of a lens system **510** as illustrated in FIG. 5. Tables 23-26 correspond to an example embodiment of a lens system **610** as illustrated in FIG. 6. Tables 27-30 correspond to an example embodiment of a lens system **710** as illustrated in FIGS. 7A and 7B. Tables 31-34 correspond to an example embodiment of a lens system **810** as illustrated in FIGS. 8A and 8B. Tables 35-38 correspond to an example embodiment of a lens system **910** as illustrated in FIG. 9.

[0093] In the Tables, all dimensions are in millimeters (mm) unless otherwise specified. L1, L2, and L3 stand for refractive lenses 1, 2, and 3, respectively. "S#" stands for surface number. A positive radius indicates that the center of curvature is to the right (object side) of the surface. A negative radius indicates that the center of curvature is to the left (image side) of the surface. "INF" stands for infinity (as used in optics). The thickness (or separation) is the axial distance to the next surface. FNO stands for F-number of the lens system. FOV stands for full field of view. $f_{35\text{ mm}}$ is the 35 mm equivalent focal length of the lens system. V_1 is the Abbe number of the first lens element, and V_2 is the Abbe number of the second lens element. Both f and EFL stand for effective focal length of the lens system, f1 stands for focal length of the first lens element, and f2 stands for focal length of the second lens element. R3f is radius of curvature of the object-side surface of lens 3, and R3r is radius of curvature of the image side surface of lens 3. Z stands for Z-height of the lens system as defined from the front (image side) vertex of the lens system to the rear vertex of the folding element (e.g., prism), as shown in FIGS. 1A and 9. REFL represents a reflective surface.

[0094] For the materials of the lens elements and IR filter, a refractive index N_d at the helium d-line wavelength is

provided, as well as an Abbe number V_d relative to the d-line and the C- and F-lines of hydrogen. The Abbe number, V_d , may be defined by the equation:

$$V_d = (N_d - 1) / (N_F - N_C),$$

where N_F and N_C are the refractive index values of the material at the F and C lines of hydrogen, respectively.

[0095] Referring to the Tables of aspheric coefficients (Tables, 2, 7, 11, 16, 20, 24, 28, 32, and 36), the aspheric equation describing an aspherical surface may be given by:

$$Z = (cr^2 / (1 + \sqrt{1 - (1 + K)c^2 r^2})) + A_4 r^4 + A_6 r^6 + A_8 r^8 + A_{10} r^{10} + A_{12} r^{12} + A_{14} r^{14} + A_{16} r^{16} + A_{18} r^{18} + A_{20} r^{20}$$

where Z is the sag of surface parallel to the z-axis (the z-axis and the optical axis are coincident in these example embodiments), r is the radial distance from the vertex, c is the curvature at the pole or vertex of the surface (the reciprocal of the radius of curvature of the surface), K is the conic constant, and A_4 - A_{20} are the aspheric coefficients. In the Tables, "E" denotes the exponential notation (powers of 10).

[0096] Note that the values given in the following Tables for the various parameters in the various embodiments of the lens system are given by way of example and are not intended to be limiting. For example, one or more of the parameters for one or more of the surfaces of one or more of the lens elements in the example embodiments, as well as parameters for the materials of which the elements are composed, may be given different values while still providing similar performance for the lens system. In particular, note that some values in the Tables may be scaled up or down for larger or smaller implementations of a camera using an embodiment of a lens system as described herein.

[0097] Further note that surface numbers (S#) of the elements in the various embodiments of the lens system as shown in the Tables are listed from a first surface 0 at the object plane to a last surface at the image plane/photosensor surface. FIGS. 10A and 10B illustrate numbering of the surfaces as used in the Tables. As shown in FIG. 10A, in some embodiments of a compact folded lens system as described herein, the image side surface of the first lens element (lens 1) may be flat/plano (e.g., lens 1 may be plano-convex), and the image side surface of lens 1 may be at/in contact with the object side surface of the light folding

prism 40 to effectively form a single combined unit or element. In these embodiments, the image side surface of lens 1 and the object side surface of the prism 40 form and are designated as a single surface, and the surfaces are numbered as illustrated in FIG. 10A:

- [0098]** S0—Object plane
- [0099]** S1—Aperture stop
- [0100]** S2—Lens 1, object side surface
- [0101]** S3—Prism 40, image side surface
- [0102]** S4—Prism 40, reflective surface
- [0103]** S5—Prism, object side surface
- [0104]** S6—Lens 2, object side surface
- [0105]** S7—Lens 2, image side surface
- [0106]** S8—Lens 3, object side surface
- [0107]** S9—Lens 3, image side surface
- [0108]** S10—IR filter 50, object side surface
- [0109]** S11—IR filter 50, image side surface
- [0110]** S12—Photosensor 20, image plane

[0111] However, in some embodiments, as shown in FIG. 10B and in Tables 10-14 corresponding to an embodiment of a lens system 310 as illustrated in FIGS. 3A and 3B, the image side surface of lens 1 may be convex, concave, or flat-plano, and lens 1 and the folding element (prism) may be air-spaced. In these embodiments, the image side surface of lens 1 and the object side surface of the prism 40 are designated as separate surfaces, and the surfaces are numbered as illustrated in FIG. 10B:

- [0112]** S0—Object plane
- [0113]** S1—Aperture stop
- [0114]** S2—Lens 1, object side surface
- [0115]** S3—Lens 1, image side surface
- [0116]** S4—Prism 40, image side surface
- [0117]** S5—Prism 40, reflective surface
- [0118]** S6—Prism, object side surface
- [0119]** S7—Lens 2, object side surface
- [0120]** S8—Lens 2, image side surface
- [0121]** S9—Lens 3, object side surface
- [0122]** S10—Lens 3, image side surface
- [0123]** S11—IR filter 50, object side surface
- [0124]** S12—IR filter 50, image side surface
- [0125]** S13—Photosensor 20, image plane

TABLE 1

Lens system 110					
Fno = 2.6, EFL = 12 mm, FOV = 23.5°, f _{35mm} = 103 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N_d	Abbe Number V_d
Object	0	INF	INF **1		
Stop	1	INF	-0.559		
L1	*2	4.79	0.791	1.513	56.6
Prism	3	INF	2.503	1.755	27.6
	Decenter (1)				
	4	INF	-2.503	REFL	
	Bend (1)				
	5	INF	-0.4		
L2	*6	12.618	-1.1	1.651	21.5
	*7	-15.437	-2.0314		
L3	*8	-2.663	-1.31	1.545	55.9
	*9	-2.889	-3.534		
IR filter	10	INF	-0.21	1.517	64.2
	11	INF	-0.1		
Sensor	12	INF	0 **2		

*Annotates aspheric surfaces (aspheric coefficients given in Table 2)

**Annotates zoom parameters (values given in Table 4)

TABLE 2

Aspheric Coefficients (Lens System 110)					
K	Surface (S#)				
	S2 0	S6 0	S7 0	S8 0	S9 0
A4	-1.32813E-04	6.25381E-03	1.16388E-02	1.57550E-02	1.08702E-02
A6	-1.18004E-05	-1.75760E-03	-3.19460E-03	5.84137E-04	1.17471E-03
A8	0.00000E+00	2.52318E-04	5.75871E-04	-4.73641E-05	7.75787E-05
A10	0.00000E+00	-3.09906E-05	-1.13692E-04	4.04236E-05	-8.08033E-05
A12	0.00000E+00	3.51844E-06	1.70836E-05	-7.61907E-06	1.85788E-05
A14	0.00000E+00	-1.71816E-07	-1.13786E-06	8.24266E-07	-1.31203E-06
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 3

Decentering Constants (Lens System 110)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 4

Zoom Parameters (Lens System 110)		
**Zoom parameters	Position-1	Position-2
**1	Infinity	500 mm
**2	0.000	-0.294 mm

TABLE 5

Optical Definitions (Lens system 110)			
EFL	12 mm	V ₁	56.6
FNO	2.6	V ₂	21.5
FOV	23.5°	f/f ₁	0.74
f _{35mm}	103 mm	f/f ₂	1.57
Z	5.8 mm	R3f/R3r	0.93

TABLE 6

Lens system 210					
Fno = 2.1, EFL = 10 mm, FOV = 25.1°, f _{35mm} = 95 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N _d	Abbe Number V _d
Object	0	INF	INF		
Stop	1	INF	-0.7049		
L1	*2	4.1908	0.9349	1.513	56.6
Prism	3	INF	2.5361	1.755	27.6
	Decenter (1)				
	4	INF	-2.5361	REFL	
	Bend (1)				
	5	INF	-0.2401		
L2	*6	-23.2221	-1.1000	1.651	21.5
	*7	-3.9100	-1.6457		
L3	*8	-2.5271	-1.3100	1.545	55.9
	*9	-3.3298	-2.0649		
IR filter	10	INF	-0.2100	1.517	64.2
	11	INF	-0.1000		
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 7)

TABLE 7

Aspheric Coefficients (Lens System 210)					
K	Surface (S#)				
	S2 0	S6 0	S7 0	S8 0	S9 0
A4	-2.27630E-04	1.66965E-02	2.17812E-02	9.21295E-03	-4.36565E-04
A6	-2.38687E-05	-8.53386E-03	-6.71712E-03	5.98333E-03	4.11661E-03
A8	0.00000E+00	6.32611E-03	1.95176E-03	-6.02221E-03	-2.47270E-03
A10	0.00000E+00	-3.37289E-03	-4.87582E-04	3.45128E-03	5.98747E-04
A12	0.00000E+00	1.10753E-03	6.23350E-05	-1.16209E-03	5.97803E-05
A14	0.00000E+00	-2.07907E-04	-1.42163E-06	2.30826E-04	-6.67851E-05
A16	0.00000E+00	1.95047E-05	0.00000E+00	-2.50104E-05	1.26669E-05
A18	0.00000E+00	-6.02309E-07	0.00000E+00	0.00000E+00	-7.50902E-07
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 8

Decentering Constants (Lens System 210)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 9

Optical Definitions (Lens system 210)			
EFL	10 mm	V ₁	56.6
FNO	2.1	V ₂	21.5
FOV	25.1°	f/f ₁	0.70
f _{35mm}	95 mm	f/f ₂	1.95
Z	6 mm	R3f/R3r	0.76

TABLE 10

Lens system 310					
Fno = 3.3, EFL = 14 mm, FOV = 20.3°, f _{35mm} = 120 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N _d	Abbe Number V _d
Object	0	INF	INF **1		
Stop	1	INF	-0.1		
L1	*2	6.2947	0.6421	1.513	56.6
	3	-26.5633	0.1		
Prism	4	INF	2.4188	1.755	27.6
	Decenter (1)				
Lens 2	5	INF	-2.4188	REFL	
	Bend (1)				
	6	INF	-0.4		
L2	*7	4.3234	-1.1	1.651	21.5
	*8	10	-1.5007		
L3	*9	-2.9320	-1.31	1.545	55.9
	*10	-3.0122	-5.9174		
IR filter	11	INF	-0.21	1.517	64.2
	12	INF	-0.1		
Sensor	13	INF	0 **2		

*Annotates aspheric surfaces (aspheric coefficients given in Table 11)

**Annotates zoom parameters (values given in Table 13)

TABLE 11

Aspheric Coefficients (Lens System 310)					
K	Surface (S#)				
	S2 0	S7 0	S8 0	S9 0	S10 0
A4	-7.53594E-04	-2.05740E-02	-1.01611E-02	1.57282E-02	1.57571E-02
A6	-5.97512E-05	5.46667E-04	-1.40041E-03	-8.47553E-04	-6.73362E-04
A8	0.00000E+00	1.06822E-04	2.50611E-04	1.39986E-04	9.71331E-05
A10	0.00000E+00	-3.80956E-05	-2.59255E-05	-5.92396E-06	-4.36111E-06
A12	0.00000E+00	4.98149E-06	-1.48356E-06	3.29881E-07	1.58201E-06
A14	0.00000E+00	-2.27653E-07	3.81073E-07	7.35242E-09	-2.16510E-07
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 11-continued

Aspheric Coefficients (Lens System 310)					
K	Surface (S#)				
	S2	S7	S8	S9	S10
	0	0	0	0	0
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 12

Decentering Constants (Lens System 310)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 13

Zoom Parameters (Lens System 310)			
**Zoom parameters	Position-1	Position-2	
**1	Infinity	400 mm	
**2	0.000	-0.509 mm	

TABLE 14

Optical Definitions (Lens system 310)			
EFL	14 mm	V ₁	56.6
FNO	3.3	V ₂	21.5
FOV	20.3°	f/f1	1.41
f _{35mm}	120 mm	f/f2	0.79
Z	5.6 mm	R3f/R3r	0.97

TABLE 15

Lens system 410 Fno = 2.4, EFL = 11 mm, FOV = 22.9°, f _{35mm} = 105 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N _d	Abbe Number V _d
Object	0	INF	INF		
Stop	1	INF	-0.6529		
L1	*2	4.1356	0.8829	1.545	56
Prism	3	INF	2.4771	1.651	21.5
Lens 2	Decenter (1)				
	4	INF	-2.4771	REFL	
Bend (1)	5	INF	-0.6921		
	*6	2.6294	-0.6000		
L2	*7	8.7797	-0.7258	1.651	21.5
	*8	-3.0194	-1.0417		
	*9	-4.4738	-3.0988	1.585	29.9
IR filter	10	INF	-0.2100		
	11	INF	-0.1000	1.517	64.2
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 16)

TABLE 16

Aspheric Coefficients (Lens System 410)					
K	Surface (S#)				
	S2	S6	S7	S8	S9
	0	0	0	0	0
A4	-3.27067E-04	-3.77803E-02	-9.36581E-03	3.64492E-02	1.98035E-02
A6	-3.44231E-05	2.77969E-03	-7.36285E-03	-6.81817E-03	-9.64138E-04
A8	0.00000E+00	-1.25370E-03	1.49418E-03	1.21923E-03	-3.31076E-04
A10	0.00000E+00	5.73173E-04	-2.45312E-04	-3.00080E-04	2.93677E-06
A12	0.00000E+00	-1.14250E-04	4.74314E-05	6.16689E-05	2.42630E-05
A14	0.00000E+00	8.90788E-06	-2.64123E-06	-4.64269E-06	-2.84035E-06
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 16-continued

Aspheric Coefficients (Lens System 410)					
K	Surface (S#)				
	S2	S6	S7	S8	S9
	0	0	0	0	0
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 17

Decentering Constants (Lens System 410)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 18

Optical Definitions (Lens system 410)			
EFL	11 mm	V ₁	56.6
FNO	2.4	V ₂	21.5
FOV	22.9°	f/f1	0.88
f _{35mm}	105 mm	f/f2	2.1
Z	5.85 mm	R3f/R3r	0.92

TABLE 19

Lens system 510					
Fno = 3.2, EFL = 12 mm, FOV = 28.5°, f _{35mm} = 84 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N _d	Abbe Number V _d
Object	0	INF	INF		
Stop	1	INF	-0.3526		
L1	*2	4.7591	0.5826	1.513	56.6
Prism	3	INF	2.1303	1.755	27.6
Decenter (1)					
Lens 2	4	INF	-2.1303	REFL	
Bend (1)					
	5	INF	-0.4000		
L2	*6	62.3900	-0.8000	1.651	21.5
	*7	-8.0000	-2.0287		
L3	*8	-2.7924	-1.3100	1.545	55.9
	*9	-3.0772	-4.2239		
IR filter	10	INF	-0.2100	1.517	64.2
	11	INF	-0.1000		
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 20)

TABLE 20

Aspheric Coefficients (Lens System 510)					
	Surface (S#)				
	S2	S6	S7	S8	S9
K	0	0	0	0	0
A4	-2.95143E-04	6.41822E-03	1.24501E-02	1.56218E-02	1.13647E-02
A6	-4.15865E-05	-3.07344E-03	-4.81181E-03	1.95660E-04	4.84332E-04
A8	0.00000E+00	4.98332E-04	9.38776E-04	4.58619E-05	1.97393E-04
A10	0.00000E+00	-2.84778E-05	-1.63568E-04	9.76688E-06	-7.89325E-05
A12	0.00000E+00	-8.82972E-06	1.59358E-05	-3.11560E-06	1.21222E-05
A14	0.00000E+00	1.63394E-06	-2.82729E-07	4.01709E-07	-6.35702E-07
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 21

Decentering Constants (Lens System 510)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 22

Optical Definitions (Lens system 510)			
EFL	12 mm	V_1	56.6
FNO	3.2	V_2	21.5
FOV	28.5°	f/f1	0.74
f_{35mm}	84 mm	f/f2	1.51
Z	4.85 mm	R3f/R3r	0.91

TABLE 23

Lens system 610					
Fno = 2.8, EFL = 10 mm, FOV = 28°, f_{35mm} = 86 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N_d	Abbe Number V_d
Object	0	INF	INF		
Stop	1	INF	-0.1000		
L1	*2	4.3054	0.6229	1.513	56.6
Prism	3	INF	2.0641	1.755	27.6
	Decenter (1)				
Lens 2	4	INF	-2.0641	REFL	
	Bend (1)				
	5	INF	-0.4000		
L2	*6	13.1995	-1.1000	1.651	21.5
	*7	-10.2750	-1.4081		
L3	*8	-2.3481	-1.3100	1.545	55.9
	*9	-2.8574	-3.2429		
IR filter	10	INF	-0.2100	1.517	64.2
	11	INF	-0.1000		
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 24)

TABLE 24

Aspheric Coefficients (Lens System 610)					
Surface (S#)					
	S2	S6	S7	S8	S9
K	0	0	0	0	0
A4	-1.67061E-04	1.31861E-02	2.90682E-02	2.93473E-02	1.30616E-02
A6	-1.28950E-05	-3.36712E-03	-7.93268E-03	1.09078E-03	3.83832E-03
A8	0.00000E+00	2.96525E-04	1.72439E-03	1.35728E-04	-2.21086E-04
A10	0.00000E+00	1.78541E-04	-2.64817E-04	8.13319E-05	-1.42155E-04
A12	0.00000E+00	-7.05953E-05	2.47809E-05	-2.82860E-05	4.12877E-05
A14	0.00000E+00	8.56770E-06	-7.05453E-07	4.70059E-06	-3.28441E-06

TABLE 24-continued

Aspheric Coefficients (Lens System 610)					
Surface (S#)					
	S2	S6	S7	S8	S9
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 25

Decentering Constants (Lens System 610)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 26

Optical Definitions (Lens system 610)				
EFL	10 mm	V_1	56.6	
FNO	2.8	V_2	21.5	
FOV	28°	f/f1	0.68	
f_{35mm}	86 mm	f/f2	1.7	
Z	4.75 mm	R3f/R3r	0.82	

TABLE 28-continued

Aspheric Coefficients (Lens System 710)				
Surface (S#)				
	S2	S6	S8	S9
A14	0.00000E+00	8.56770E-06	6.90588E-07	1.34154E-06
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 27

Lens system 710 Fno = 3.8, EFL = 16 mm, FOV = 17.8°, f_{35mm} = 137 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N_d	Abbe Number V_d
Object	0	INF	INF		
Stop	1	INF	-0.3655		
L1	*2	5.5666	0.6155	1.513	56.6
Prism	3	INF	2.4773	1.755	27.6
Decenter (1)					
Lens 2	4	INF	-2.4773	REFL	
Bend (1)					
	5	INF	-0.4000		
L2	*6	9.4597	-1.1000	1.651	21.5
	7	57.2358	-5.0141		
L3	*8	-3.6676	-1.3100	1.545	55.9
	*9	-3.2803	-4.1430		
IR filter	10	INF	-0.2100	1.517	64.2
	11	INF	-0.1000		
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 28)

TABLE 28

Aspheric Coefficients (Lens System 710)				
Surface (S#)				
	S2	S6	S8	S9
K	0	0	0	0
A4	-7.39550E-07	1.88716E-04	4.69962E-03	5.45999E-03
A6	1.21192E-05	4.24110E-05	9.74285E-04	1.50061E-03
A8	0.00000E+00	2.36650E-06	-3.07164E-04	-4.57976E-04
A10	0.00000E+00	0.00000E+00	9.15076E-05	1.46920E-04
A12	0.00000E+00	0.00000E+00	-1.25986E-05	-2.23759E-05

TABLE 29

Decentering Constants (Lens System 710)						
Decenter	X	Y	Z	Alpha (deg)	Beta (deg)	Gamma (deg)
D (1) and Bend (1)	0	0	0	45	0	0

TABLE 30

Optical Definitions (Lens system 710)			
EFL	16 mm	V_1	56.6
FNO	3.8	V_2	21.5
FOV	17.8°	f/f1	0.84
f_{35mm}	137 mm	f/f2	1.09
Z	5.5 mm	R3f/R3r	1.12

TABLE 34-continued

Optical Definitions (Lens system 810)			
f_{35mm}	120 mm	f/f2	1.48
Z	5.4 mm	R3f/R3r	0.87

TABLE 31

Lens system 810					
Fno = 3.2, EFL = 14 mm, FOV = 20.1°, f_{35mm} = 120 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N_d	Abbe Number V_d
Object	0	INF	INF		
Stop	1	INF	-0.3806		
L1	*2	6.0162	0.6108	1.513	56.6
Prism	3	INF	2.3889	1.755	27.6
Decenter (1)					
Lens 2	4	INF	-2.3889	REFL	
Bend (1)					
	5	INF	-0.4000		
L2	*6	10.0838	-1.5000	1.651	21.5
	7	-41.7944	-2.9678		
L3	*8	-3.0676	-2.1000	1.545	55.9
	*9	-3.5230	-4.7965		
IR filter	10	INF	-0.2100	1.517	64.2
	11	INF	-0.1000		
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 32)

TABLE 32

Aspheric Coefficients (Lens System 810)				
Surface (S#)				
	S2	S6	S8	S9
K	0	0	0	0
A4	-1.37180E-04	-1.27420E-03	2.45927E-03	-3.53430E-03
A6	7.12659E-07	6.01099E-05	6.23315E-04	7.23182E-04
A8	0.00000E+00	5.40605E-07	-1.94068E-04	-3.28264E-04
A10	0.00000E+00	0.00000E+00	5.29622E-05	8.55874E-05
A12	0.00000E+00	0.00000E+00	-6.33171E-06	-9.64805E-06
A14	0.00000E+00	0.00000E+00	3.40851E-07	4.35434E-07
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 33

Decentering Constants (Lens System 810)						
Decenter	X	Y	Z	Alpha(deg)	Beta(deg)	Gamma(deg)
D(1) and Bend (1)	0	0	0	45	0	0

TABLE 34

Optical Definitions (Lens system 810)			
EFL	14 mm	V_1	56.6
FNO	3.2	V_2	21.5
FOV	20.1°	f/f1	0.77

TABLE 35

Lens system 910					
Fno = 3.0, EFL = 12 mm, FOV = 23.5°, f_{35mm} = 103 mm					
Element	Surface (S#)	Radius (mm)	Thickness or separation (mm)	Refractive Index N_d	Abbe Number V_d
Object	0	INF	INF		
Stop	1	INF	-0.4129		
L1	*2	6.0162	0.6429	1.513	56.6
Prism	3	INF	2.2006	1.755	27.6
Decenter (1)					
Lens 2	4	INF	-2.2006	REFL	
Bend (1)					
	5	INF	-0.4000		
L2	*6	9.7361	-1.1000	1.651	21.5
	*7	-22.7209	-1.6487		
L3	*8	-2.5469	-1.3100	1.545	55.9
	*9	-2.7328	-4.2993		
IR filter	10	INF	-0.2100	1.517	64.2
	11	INF	-0.1000		
Sensor	12	INF	0.0000		

*Annotates aspheric surfaces (aspheric coefficients given in Table 36)

TABLE 36

Aspheric Coefficients (Lens System 910)				
	Surface (S#)			
	S2	S6	S8	S9
K	0	0	0	0
A4	-1.63365E-04	6.58456E-03	1.70894E-02	2.48425E-02
A6	-2.42913E-05	-2.57282E-03	-5.08885E-03	3.99244E-04
A8	0.00000E+00	3.78108E-04	9.27077E-04	4.41383E-05
A10	0.00000E+00	-1.20871E-05	-1.23550E-04	2.69972E-05
A12	0.00000E+00	-9.72383E-06	6.20626E-06	-1.02741E-05
A14	0.00000E+00	1.96232E-06	8.41423E-07	1.93889E-06
A16	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A18	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00
A20	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00

TABLE 37

Decentering Constants (Lens System 910)						
Decenter	X	Y	Z	Alpha(deg)	Beta(deg)	Gamma (deg)
D(1) and Bend (1)	0	0	0	45	0	0

TABLE 38

Optical Definitions (Lens system 910)			
EFL	12 mm	V ₁	56.6
FNO	3.0	V ₂	21.5
FOV	23.5°	f/f ₁	0.74
f _{35mm}	103 mm	f/f ₂	1.58
Z	5.45 mm	R3f/R3r	0.93

Example Flowchart

[0126] FIG. 11 is a high-level flowchart of a method for capturing images using a camera with a lens system that includes three lens elements and a folding element as illustrated in FIGS. 1 through 10B, according to some embodiments. As indicated at 2400, light from an object field in front of the camera is received at a first lens element of the camera through an aperture stop. In some embodiments, the aperture stop may be located at the first lens element and behind the front vertex of the lens system. As indicated at 2402, the first lens element refracts the light on a first axis AX1 to a light folding element such as a prism. As indicated at 2404, the light is redirected by the folding element to a second lens element on a second axis AX2. As indicated at 2406, the light is then refracted by the second lens element to a third lens element on the second axis AX2. As indicated at 2408, the light is then refracted by the third lens element to form an image at an image plane at or near the surface of a photosensor. As indicated at 2414, the image is captured by the photosensor. While not shown, in some embodiments, the light may pass through an infrared filter that may for example be located between the third lens element and the photosensor.

[0127] In some embodiments, the elements referred to in FIG. 11 may be configured as illustrated in any of FIGS. 1 through 10B. However, note that variations on the example as given in the Figures are possible while achieving similar optical results.

Example Computing Device

[0128] FIG. 12 illustrates an example computing device, referred to as computer system 4000, that may include or host embodiments of the camera as illustrated in FIGS. 1 through 11. In addition, computer system 4000 may implement methods for controlling operations of the camera and/or for performing image processing of images captured with the camera. In different embodiments, computer system 4000 may be any of various types of devices, including, but not limited to, a personal computer system, desktop computer, laptop, notebook, tablet or pad device, slate, or netbook computer, mainframe computer system, handheld computer, workstation, network computer, a camera, a set top box, a mobile device, a mobile multipurpose device, a wireless phone, a smartphone, a consumer device, video game console, handheld video game device, application server, storage device, a television, a video recording device, or in general any type of computing or electronic device.

[0129] In the illustrated embodiment, computer system 4000 includes one or more processors 4010 coupled to a system memory 4020 via an input/output (I/O) interface 4030. Computer system 4000 further includes a network interface 4040 coupled to I/O interface 4030, and one or more input/output devices 4050, such as cursor control device 4060, keyboard 4070, and display(s) 4080. Computer system 4000 may also include one or more cameras 4090, for example one or more cameras as described above with respect to FIGS. 1 through 11, which may also be coupled to I/O interface 4030, or one or more cameras as described above with respect to FIGS. 1 through 11 along with one or more other cameras such as wide-field cameras.

[0130] In various embodiments, computer system 4000 may be a uniprocessor system including one processor 4010, or a multiprocessor system including several processors 4010 (e.g., two, four, eight, or another suitable number). Processors 4010 may be any suitable processor capable of executing instructions. For example, in various embodiments processors 4010 may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors 4010 may commonly, but not necessarily, implement the same ISA.

[0131] System memory 4020 may be configured to store program instructions 4022 and/or data 4032 accessible by processor 4010. In various embodiments, system memory 4020 may be implemented using any suitable memory technology, such as static random access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. In the illustrated embodiment, program instructions 4022 may be configured to implement various interfaces, methods and/or data for controlling operations of camera 4090 and for capturing and processing images with integrated camera 4090 or other methods or data, for example interfaces and methods for capturing, displaying, processing, and storing images captured with camera 4090. In some embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media or on similar media separate from system memory 4020 or computer system 4000.

[0132] In one embodiment, I/O interface 4030 may be configured to coordinate I/O traffic between processor 4010, system memory 4020, and any peripheral devices in the

device, including network interface **4040** or other peripheral interfaces, such as input/output devices **4050**. In some embodiments, I/O interface **4030** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **4020**) into a format suitable for use by another component (e.g., processor **4010**). In some embodiments, I/O interface **4030** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **4030** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **4030**, such as an interface to system memory **4020**, may be incorporated directly into processor **4010**.

[0133] Network interface **4040** may be configured to allow data to be exchanged between computer system **4000** and other devices attached to a network **4085** (e.g., carrier or agent devices) or between nodes of computer system **4000**. Network **4085** may in various embodiments include one or more networks including but not limited to Local Area Networks (LANs) (e.g., an Ethernet or corporate network), Wide Area Networks (WANs) (e.g., the Internet), wireless data networks, some other electronic data network, or some combination thereof. In various embodiments, network interface **4040** may support communication via wired or wireless general data networks, such as any suitable type of Ethernet network, for example; via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks; via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

[0134] Input/output devices **4050** may, in some embodiments, include one or more display terminals, keyboards, keypads, touchpads, scanning devices, voice or optical recognition devices, or any other devices suitable for entering or accessing data by computer system **4000**. Multiple input/output devices **4050** may be present in computer system **4000** or may be distributed on various nodes of computer system **4000**. In some embodiments, similar input/output devices may be separate from computer system **4000** and may interact with one or more nodes of computer system **4000** through a wired or wireless connection, such as over network interface **4040**.

[0135] As shown in FIG. 12, memory **4020** may include program instructions **4022**, which may be processor-executable to implement any element or action to support integrated camera **4090**, including but not limited to image processing software and interface software for controlling camera **4090**. In some embodiments, images captured by camera **4090** may be stored to memory **4020**. In addition, metadata for images captured by camera **4090** may be stored to memory **4020**.

[0136] Those skilled in the art will appreciate that computer system **4000** is merely illustrative and is not intended to limit the scope of embodiments. In particular, the computer system and devices may include any combination of hardware or software that can perform the indicated functions, including computers, network devices, Internet appliances, PDAs, wireless phones, pagers, video or still cameras, etc. Computer system **4000** may also be connected to other devices that are not illustrated, or instead may operate

as a stand-alone system. In addition, the functionality provided by the illustrated components may in some embodiments be combined in fewer components or distributed in additional components. Similarly, in some embodiments, the functionality of some of the illustrated components may not be provided and/or other additional functionality may be available.

[0137] Those skilled in the art will also appreciate that, while various items are illustrated as being stored in memory or on storage while being used, these items or portions of them may be transferred between memory and other storage devices for purposes of memory management and data integrity. Alternatively, in other embodiments some or all of the software components may execute in memory on another device and communicate with the illustrated computer system **4000** via inter-computer communication. Some or all of the system components or data structures may also be stored (e.g., as instructions or structured data) on a computer-accessible medium or a portable article to be read by an appropriate drive, various examples of which are described above. In some embodiments, instructions stored on a computer-accessible medium separate from computer system **4000** may be transmitted to computer system **4000** via transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link. Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include a non-transitory, computer-readable storage medium or memory medium such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g. SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc. In some embodiments, a computer-accessible medium may include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

[0138] The methods described herein may be implemented in software, hardware, or a combination thereof, in different embodiments. In addition, the order of the blocks of the methods may be changed, and various elements may be added, reordered, combined, omitted, modified, etc. Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. The various embodiments described herein are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described herein as a single instance. Boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of claims that follow. Finally, structures and functionality presented as discrete components in the example configurations may be implemented as a combined structure or component. These and other variations, modifications, additions, and improvements may fall within the scope of embodiments as defined in the claims that follow.

What is claimed is:

1. A lens system, comprising:

- a plurality of elements arranged along a folded optical axis of the lens system, wherein the plurality of elements includes, in order along the folded optical axis from an object side to an image side of the lens system:
 - a first lens element on a first portion of the folded optical axis having a convex object-side surface in the paraxial region;
 - a light folding element configured to redirect light from the first lens element to a second portion of the folded optical axis;
 - a second lens element on the second portion of the folded optical axis; and
 - a third lens element on the second portion of the folded optical axis having a concave image-side surface in the paraxial region.

2. The lens system as recited in claim 1, wherein the first lens element has positive refractive power.

3. The lens system as recited in claim 1, wherein the third lens element has a convex object-side surface in the paraxial region.

4. The lens system as recited in claim 1, wherein the lens system further comprises an aperture stop located between the object side of the lens system and the light folding element.

5. The lens system as recited in claim 1, wherein the lens system provides a 35 mm equivalent focal length in the range of 80-200 millimeters and less than 6.5 millimeters of Z-height measured from a front vertex of the lens system to a rear vertex of the folding element.

6. The lens system as recited in claim 1, wherein the lens system provides a 35 mm equivalent focal length in the range of 85-200 millimeters and less than 6 millimeters of Z-height measured from a front vertex of the lens system to a rear vertex of the folding element.

7. The lens system as recited in claim 1, wherein the first lens element is formed of an optical material with Abbe number $V_d > 45$, and the second lens element is formed of an optical material with Abbe number $V_d < 35$.

8. The lens system as recited in claim 1, wherein the first lens element is formed of an optical material with Abbe number $V_d > 40$, and the second lens element is formed of an optical material with Abbe number $V_d < 30$.

9. The lens system as recited in claim 1, wherein the lens system satisfies one or more of the relationships:

$$0.5 < |f/f_1| < 2$$

$$0.4 < |f/f_2| < 2.5$$

$$0.5 < |R_3/R_{3r}| < 1.5$$

where f is effective focal length of the lens system, f_1 is focal length of the first lens element, f_2 is focal length of the second lens element, R_3 is radius of curvature of the object-side surface of the third lens element, and R_{3r} is radius of curvature of the image side surface of the third lens element.

10. The lens system as recited in claim 1, wherein at least one surface of at least one of the plurality of lens elements is aspheric.

11. The lens system as recited in claim 1, wherein at least one of the lens elements is formed of lightweight polymer or plastic material.

12. The lens system as recited in claim 1, wherein the light folding element is a prism.

13. The lens system as recited in claim 12, wherein an image side surface of the first lens element is flat/plano, and wherein the image side surface of the first lens element is in contact with the object side surface of the prism.

14. The lens system as recited in claim 12, wherein an image side surface of the first lens element is convex, concave, or flat/plano, and wherein the image side surface of the first lens element is not in contact with the object side surface of the prism.

15. The lens system as recited in claim 1, wherein effective focal length of the lens system is within a range of 10 millimeters to 16 millimeters.

16. A camera, comprising:

a photosensor configured to capture light projected onto a surface of the photosensor; and

a folded lens system configured to refract light from an object field located in front of the camera to form an image of a scene at an image plane at or near the surface of the photosensor, wherein the lens system comprises three refractive lens elements arranged along a folded optical axis of the camera from an object side to an image side and a light folding element located between a first and second lens element from the object side and configured to redirect light from a first axis onto a second axis;

wherein the folded lens system provides a 35 mm equivalent focal length in the range of 80-200 millimeters and 6.5 millimeters or less of Z-height measured from a front vertex of the lens system to a rear vertex of the folding element.

17. The camera as recited in claim 16, wherein effective focal length of the lens system is within a range of 10 millimeters to 16 millimeters, and wherein the photosensor is between 4 millimeters and 8 millimeters in a diagonal dimension.

18. The camera as recited in claim 16, wherein the photosensor is configured to move on one or more axes relative to the lens system to adjust focus of the camera.

19. A device, comprising:

one or more processors;

one or more cameras; and

a memory comprising program instructions executable by at least one of the one or more processors to control operations of the one or more cameras;

wherein at least one of the one or more cameras is a camera comprising:

a photosensor configured to capture light projected onto a surface of the photosensor; and

a folded lens system configured to refract light from an object field located in front of the camera to form an image of a scene at an image plane proximate to the surface of the photosensor, wherein the lens system comprises three refractive lens elements arranged along a folded optical axis of the lens system from an object side to an image side and a light folding element configured to redirect light from the first lens element on the object side to a second portion of the folded optical axis;

wherein the lens system provides a 35 mm equivalent focal length in the range of 80-200 millimeters and less

than 6.5 millimeters of Z-height measured from a front vertex of the lens system to a rear vertex of the folding element.

20. The device as recited in claim **19**, wherein the lens system further comprises at least one aperture stop located between a front vertex of the lens system and the light folding element.

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